A conditional lattice computation of the branch-bias prefactor for a gravitational-instanton cosmological-constant residual

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Abstract

We compute on the lattice the branch-bias prefactor entering the master formula for the cosmological constant, conditional on the hidden-sector action $S_{\rm inst}$ taken as external input from a companion structural analysis. The framework is the Sakharov–Volovik induced-gravity tradition extended by a coframe / teleparallel completion, with an analytic constitutive subtraction that removes polynomial ultraviolet hierarchies order by order in the one-loop branch-bias expansion. The surviving infrared residual is reopened nonperturbatively by a self-dual $SU(2)$ gravitational instanton of Harrington–Shepard type on $S^1 \times \mathbb{R}^3$, whose action supplies a dimensional-transmutation exponential and whose one-loop branch determinant supplies an $\mathcal{O}(1)$ prefactor. We compute on three independent lattices spanning a factor of 11 in volume and a factor of 1.5 in caloron scale the three lattice ingredients entering that prefactor: the physical Jacobian of collective coordinates, the phase of the instanton contribution, and the normalized branch-response bracket together with its finite-ξ Simpson correction and lattice-size-independence diagnostics. The branch-response determinant is evaluated with the adaptive-prime prescription validated internally by lattice-size stability and by an independent projected-complement cross-check. We establish a factorization proposition for the master formula, a phase theorem fixing the parity-even angle to $+1$ on every background considered, and a stability proposition for the Simpson nonlinear correction across the three production lattices. Conditional on the hidden-sector action $S_{\rm inst} \simeq 276$ supplied by a companion analysis, the assembled lattice-determined prefactor is 0.81 ± 0.03, so that

$$\Lambda_{\rm obs}/M_{\rm P}^4 \;=\; (0.81 \pm 0.03) \times 10^{-120} \qquad \text{(conditional on } S_{\rm inst} \simeq 276\text{)}.$$

1 Introduction

The cosmological constant problem is the discrepancy between the small positive vacuum energy density inferred from cosmological observations and the much larger values expected from effective field theory applied to the known matter sectors of the Standard Model and its extensions [1–4]. Reproducing the observed magnitude $\Lambda_{\rm obs} \sim 10^{-120}\, M_{\rm P}^4$ within a controlled theoretical framework, without relying on anthropic selection or on cancellations between unrelated sectors, has remained an open problem in high-energy theory. Several distinct approaches have been proposed, including supersymmetric cancellation, adjustment mechanisms based on scalar or four-form sectors, sequestering constructions, and vacuum-selection arguments; a comparison of the present paper to these approaches is given in Section 11.

We work within the Sakharov–Volovik induced-gravity tradition [5–8], in which the gravitational effective action is generated by integrating out fluctuations of a microscopic proto-field sector on an emergent metric $g_{\rm eff}$. The Newton constant and the cosmological constant are both induced by this procedure, so that their magnitudes are fixed by the microscopic content of the

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proto-field sector and by whatever cancellation or subtraction structure the framework supplies. In the specific framework of the companion structural paper [9][1], the induced gravitational action is completed by a coframe / teleparallel sector that selects a preferred FLRW foliation, and polynomial ultraviolet hierarchies in the vacuum-energy response between two constitutive branches are removed by an analytic subtraction within that sector. The result of the subtraction is a finite infrared residual, which the present paper computes. The companion paper is deposited at DOI 10.5281/zenodo.19599312 and is under journal review.

The residual is reopened nonperturbatively by a self-dual $SU(2)$ gravitational instanton of Harrington–Shepard type [10] on $S^1 \times \mathbb{R}^3$, constructed from the underlying BPST instanton [11] with the 't Hooft sign conventions [12]. The instanton action supplies a dimensional-transmutation exponential, while the one-loop branch determinant computed on the instanton background supplies the normalized branch-response factor entering the lattice prefactor. The resulting master formula for the cosmological constant,

$$\Lambda_{\rm obs} \;\sim\; M_{\rm P}^4\, e^{-8\pi^2 a_C \mathcal{I}_{\rm lat}}\, J_{\rm phys}\, [\zeta(0) - \zeta(X_*)]\, \cos\Theta_\gamma,$$

factorizes into four UV-finite quantities after the analytic subtraction. The structural derivation of this factorization and of the accompanying no-go results ruling out local scalar-only realizations of the same residual is given in [9]; the present paper takes that derivation as input.

What this paper establishes is a lattice computation of the three factors of the master formula that are intrinsic to the branch-bias determinant: the physical Jacobian $J_{\rm phys}$, the phase $\cos\Theta_\gamma$, and the normalized branch-response bracket $\mathcal{B} \equiv 1 - \zeta(X_*)/\zeta(0)$. The dimensional-transmutation exponential $e^{-S_{\rm inst}}$ is not recomputed here; it is taken as conditional input from [9]. The physical Jacobian of the collective-coordinate measure is computed in Section 4; the phase is determined by a theorem within the parity-even class in Section 5; the branch-response bracket is measured on three independent lattices via a transported-prime Simpson campaign in Section 8, with lattice-size independence established in Section 9. Three named results form the logical spine of the computation: a factorization proposition establishing UV-finiteness of each master-formula factor, a phase theorem fixing $\cos\Theta_\gamma = +1$ in the parity-even class on every background geometry considered, and a stability proposition for the Simpson nonlinear correction across three lattices spanning a factor of 11 in volume. Combined with the conditional hidden-sector input $S_{\rm inst} \simeq 276$ from [9], the lattice computation yields an assembled prefactor $\mathcal{K} = 0.81 \pm 0.03$ multiplying the dimensional-transmutation exponential 10^{-120}; the canonical assembled statement is given in Section 10.

Organization. Section 2 fixes the master formula and the framework inputs inherited from [9]. Section 3 defines the lattice discretization and admissibility window. Section 4 computes the physical Jacobian. Section 5 establishes the phase theorem. Section 6 describes the spectral structure of the lattice vector operator and the adaptive-prime prescription used in the determinant computation. Section 7 reviews the prescriptions considered and reports an independent cross-prescription measurement. Section 8 presents the transported-prime Simpson campaign and the three-row production result. Section 9 establishes lattice-size independence across the three production rows (no formal continuum extrapolation is claimed). Section 10 assembles the final result with the full error budget. Section 11 discusses the conditional scope, the coupling assumption, and the relation to existing approaches; Section 12 concludes. Six appendices collect the operator and solver conventions, the transported-prime algorithm, the infrastructure-test suite, the $\mathcal{I}_{\rm cl}$-degraded reference rows, the bracket error budget, and the sensitivity of the assembled result to the hidden-sector action.

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2 Master formula and inherited framework inputs

2.1 Framework inputs

The framework used below, derived in the companion structural paper [9] and treated here as input, has three ingredients. First, a proto-field sector whose one-loop integration yields an infrared gravitational effective action $\Gamma_{\rm grav}[g_{\rm eff}]$ in the Sakharov [5, 6]–Volovik [7, 8] induced-gravity tradition. Second, a coframe / teleparallel completion that provides the geometric branch structure on which the constitutive scalar X is defined. Third, an analytic constitutive subtraction that removes all polynomial ultraviolet terms in the one-loop branch-bias expansion, leaving a finite infrared residual. None of these ingredients is re-derived in the present paper. The purpose of the sections that follow is to compute, on the lattice, the residual left after this subtraction.

2.2 Branch-bias mechanism and master formula

The constitutive sector admits two branches, distinguished by the value $X = 0$ or $X = X_*$ of a constitutive scalar. After the analytic subtraction of [9], the difference of the one-loop determinants between the two branches is UV-finite, and the master formula for the observed cosmological constant is

$$\Lambda_{\rm obs} \;\sim\; M_{\rm P}^4\, e^{-8\pi^2 a_C \mathcal{I}_{\rm lat}}\, J_{\rm phys}\, \left[\zeta(0) - \zeta(X_*)\right]\, \cos\Theta_\gamma. \tag{1}$$

The factors are: the dimensional-transmutation exponential $e^{-S_{\rm inst}}$ with $S_{\rm inst} = 8\pi^2 a_C \mathcal{I}_{\rm lat}$, where a_C is the constitutive-sector bare coupling and $\mathcal{I}_{\rm lat}$ the normalized lattice instanton action, both as defined in [9]; the physical Jacobian of collective coordinates $J_{\rm phys}$; the phase of the instanton contribution $\cos\Theta_\gamma$; and the one-loop branch-response bracket $\mathcal{B} \equiv 1 - \zeta(X_*)/\zeta(0)$. Equivalently,

$$\zeta(0) - \zeta(X_*) \;=\; \zeta(0)\,\mathcal{B},$$

so that the bracket $\mathcal{B}$ is the normalized branch-response factor quoted throughout the paper.

Proposition 2.1 (Factorization and finiteness)**.** *After the analytic constitutive subtraction of [9], the master formula* (1) *factorizes into four UV-finite quantities. In the present paper,* $J_{\rm phys}$*,* $\cos\Theta_\gamma$*, and* $\mathcal{B} \equiv 1 - \zeta(X_*)/\zeta(0)$ *are computed directly on the lattice, whereas* $e^{-S_{\rm inst}}$ *is treated as an inherited input from the companion structural analysis.*

Proof. The finiteness of the factorization is inherited from the analytic subtraction proved in [9]. The Jacobian $J_{\rm phys}$ is the square root of a 5×5 Gram determinant of collective-coordinate tangents and is finite by construction. The phase $\cos\Theta_\gamma$ is fixed by the parity theorem of Section 5. For the branch-response factor, the same subtraction renders $\log\zeta(X) - \log\zeta(0)$ polynomial-free order by order in the ultraviolet expansion, leaving the finite residual computed in Section 8. The exponential factor $e^{-S_{\rm inst}}$ is not recomputed here; it enters as a conditional framework input. □

2.3 Conditional scope of this paper

Two conditionals are fixed in the computation that follows, and the assembled result of Section 10 is conditional on both. The hidden-sector action entering the dimensional-transmutation exponential is taken as $S_{\rm inst} \simeq 276$ from a separate analysis of the hidden sector in [9]; all numerical content of the present paper is independent of the hidden-sector algebra, $S_{\rm inst}$ enters the master formula only multiplicatively, and the sensitivity of the assembled result to its value is quantified in Appendix F. The branch-response determinant is computed using the adaptive-prime prescription of Section 6. In the present paper that prescription is supported

by the lattice-size stability tests of Section 9 and by the independent projected-complement cross-check of Section 7.2. A first-principles dressed analytic basis for the adaptive prime is deferred to separate technical work. Nothing in the numerical pipeline below depends on the hidden-sector field content beyond the single number $S_{\rm inst}$.

3 Lattice discretization of the Harrington–Shepard caloron

3.1 Seed construction and path-ordered links

The Harrington–Shepard (HS) caloron [10] is the trivial-holonomy unit-charge periodic BPST instanton on $S^1 \times \mathbb{R}^3$ [11]. In the notation used here its gauge potential is written

$$A^a_\mu(x) \;=\; \eta^{(\pm)a}_{\mu\nu}\,\partial_\nu \log \Pi(x), \qquad \Pi(x) \;=\; 1 \;+\; \frac{\pi\,\rho^2}{\beta\, r}\,\frac{\sinh(2\pi r/\beta)}{\cosh(2\pi r/\beta) \;-\; \cos(2\pi t/\beta)}, \tag{2}$$

where ρ is the caloron scale parameter, $r = |\vec{x}|$, and $\eta^{(\pm)a}_{\mu\nu}$ denotes the 't Hooft symbol of the chosen orientation [12]. Lattice links $U_{x,\mu} \in SU(2)$ are constructed by discretized path-ordered integration of (2) along each lattice edge into $n_{\rm sub} = 16$ sub-intervals, composed as an ordered product of sub-link exponentials, and unitarity-projected onto $SU(2)$ by polar decomposition; the absolute deviation $\|U^\dagger U - \mathbb{1}\|_{\max}$ stays below 10^{-15} on every production background.

3.2 Admissibility window and clover diagnostics

The quality of the discretized seed is monitored by the clover-averaged lattice instanton action

$$\mathcal{I}_{\rm cl} \;\equiv\; \frac{1}{8\pi^2}\sum_x \mathrm{Tr}\big[F^{\rm clover}_{\mu\nu}(x)\, F^{{\rm clover}\,\mu\nu}(x)\big], \tag{3}$$

where $F^{\rm clover}_{\mu\nu}$ is the symmetric average of the four plaquettes meeting at x in the (μ,ν)-plane. In the present paper, $\mathcal{I}_{\rm cl}$ is used only as a discretization-quality diagnostic for the seeded caloron background. The continuum target for a $Q = 1$ self-dual caloron is $\mathcal{I}_{\rm cl} = 1$; deviations measure the combined effect of finite-spacing and finite-volume artefacts on the seeded profile.

The target admissibility window for the primary production rows is

$$\mathcal{I}_{\rm cl} \;\in\; [\,0.98,\, 1.00\,]. \tag{4}$$

In addition, one row at $\mathcal{I}_{\rm cl} = 1.032$ is retained in the main analysis as a stress test of the window itself: its inclusion in the weighted bracket of Section 8 is designed to test whether the bracket value is sensitive to mild excursions from (4). The empirical answer, reported in Section 8, is that it is not. All three production rows also pass the independent admissibility check of [13] with max-plaquette deviations well below the threshold 0.447 (Table 2). Rows with larger departures from unity are used only as a systematics reference and are deferred to Appendix D. All production backgrounds are seeded from the unit-charge HS caloron; no separate lattice measurement of Q is used in the present paper.

Table 1 lists the three production rows used in the remainder of the paper.

Table 2 collects the standard per-row diagnostic quantities that a lattice referee audits: the clover measurement of both the action normalization $\mathcal{I}_{\rm cl}$ and the topological charge $Q_{\rm cl}$ (numerically close but not identical at the $\mathcal{O}(10^{-1})$ level, in agreement with the independently measured self-duality violation δ_+); the average plaquette, which certifies that the discretized field is everywhere in the weak-coupling regime; the admissibility check of [13], which all three rows pass; the link-construction stability under doubling the sub-interval count, which is at the 10^{-6} level; and the unitarity of every link, which is at machine precision. The m^2-regulator dependence on Row A is reported separately: halving the infrared regulator from $m^2 = 0.01$

Row	Dimensions	ρ	$\mathcal{I}_{\rm cl}$
A	$18^3 \times 12$	3.0	0.983
B	$24^3 \times 16$	3.5	0.998
C	$32^3 \times 24$	4.5	1.032

Table 1: Production rows. Rows A and B lie inside the target admissibility window (4); Row C is retained as an explicit stress test slightly outside it, extending the range to a factor of 11 in lattice volume and a factor of 1.5 in caloron scale.

Row	$\mathcal{I}_{\rm cl}$	$Q_{\rm cl}$	δ_+	$\langle P \rangle$	$n_{\rm sub}$-stab.	$\lVert U^\dagger U - \mathbb{1} \rVert_{\rm max}$	adm.
A	0.9835	0.8942	4.54×10^{-2}	0.99995	4.2×10^{-6}	1.3×10^{-15}	pass
B	0.9982	0.9250	3.66×10^{-2}	0.99998	2.3×10^{-6}	1.3×10^{-15}	pass
C	1.0321	0.9558	3.70×10^{-2}	1.00000	1.1×10^{-6}	1.3×10^{-15}	pass

Table 2: Per-row lattice diagnostics, all read directly from the production pipeline (reproducibility archive, `scripts/lattice_diagnostics_rowsABC.py`). $\mathcal{I}_{\rm cl}$ and $Q_{\rm cl}$ are the clover-action and clover-topological-charge measurements respectively; both target the continuum value +1 for a unit-charge self-dual seed. The gap $\mathcal{I}_{\rm cl} - Q_{\rm cl}$ and the self-duality violation $\delta_+ = \sum_x |F - \tilde{F}|^2 / (4 \sum_x |F|^2)$ are the expected $\mathcal{O}(a^2)$ discretization effects of the clover field strength on a finite lattice, and decrease (as a percentage of $\mathcal{I}_{\rm cl}$) from Row A to Row C as the lattice spacing decreases relative to the caloron size ρ. The average plaquette $\langle P \rangle = (1/6N_{\rm sites}) \sum_{x,\mu<\nu} \frac{1}{2} \operatorname{Tr}(\mathbb{1} - U_{\mu\nu})$ differs from unity by at most 5×10^{-5} on all rows. The $n_{\rm sub}$-stability column reports the fractional change of $\mathcal{I}_{\rm cl}$ under doubling the link sub-interval count from $n_{\rm sub} = 16$ (production) to $n_{\rm sub} = 32$; all values are below 5×10^{-6}, confirming that the link construction is converged at production precision. The admissibility column reports the plaquette-bound admissibility check (threshold = 0.447) of [13]; all three production rows pass.

to $m^2 = 0.005$ produces a fractional change of 9.6×10^{-4} in the first-order trace response $\operatorname{Tr}[\Delta_1^{-1} K_0]$ at $\xi = 1$, confirming that the $\delta_\xi \log \zeta$ values reported in Section 8 are not sensitive to the regulator choice at the precision quoted.

3.3 Reference conventions

Throughout the production pipeline the infrared regulator is $m^2 = 0.01$ and the disformal branch parameter is $\epsilon = 0.09$, both held fixed across every row. Adjoint-vector fields are stored under the linear layout $v[\,12x + 3\mu + a\,]$ on $12V$ entries, $V = L_t L_s^3$, and the three operators entering the branch-response determinant,

$$\Delta_1 \;=\; -D^2 + 2F + m^2 \mathbb{1}, \qquad \Delta_0 \;=\; -D^2 + m^2 \mathbb{1}, \qquad K_0 \;=\; -D_0^2, \tag{5}$$

are implemented with the sign and regulator conventions fixed in Appendix A. Infrastructure self-tests verifying consistency of the code paths entering (5) are summarized in Appendix C. The computation is carried out under the parity-even hypotheses stated in Section 5. No further sectoral structure enters the numerical pipeline below.

4 Physical Jacobian

4.1 Left-trivialized tangents and the physical 5-manifold

The HS caloron of Section 3 is characterized by five collective coordinates: four translations z^μ ($\mu = 0, 1, 2, 3$) and one scale ρ. The corresponding tangents on the link manifold are constructed

in the left-trivialized frame, in which the variation of a link $U_{x,\mu}$ is written as an element of the Lie algebra $\mathfrak{su}(2)$ at the origin of the group via $\delta U_{x,\mu} = U_{x,\mu}\, t_{x,\mu}$, so that $t = \{t^a_{x,\mu}\}$ lives in the same adjoint-vector space $\mathbb{C}^{12V}$ as the operators of Section 3. Each collective-coordinate tangent is then a finite-difference derivative of the HS seed along the corresponding direction:

$$t^{(\mathrm{tr})}_\mu \;\equiv\; \frac{\mathrm{LT}\big[U(z+\delta\hat{e}_\mu)\big] - \mathrm{LT}\big[U(z-\delta\hat{e}_\mu)\big]}{2\,\delta}, \qquad t^{(\mathrm{sc})} \;\equiv\; \frac{\mathrm{LT}\big[U(\rho+\delta)\big] - \mathrm{LT}\big[U(\rho-\delta)\big]}{2\,\delta}, \tag{6}$$

with $\mathrm{LT}[\,\cdot\,]$ the left-trivialization map defined in [9] and finite-difference steps $\delta = 0.1$ for translations and $\delta = 0.05$ for scale. The five-dimensional subspace $V_{\mathrm{mod}} = \mathrm{span}\{\, t^{(\mathrm{tr})}_0, t^{(\mathrm{tr})}_1, t^{(\mathrm{tr})}_2, t^{(\mathrm{tr})}_3, t^{(\mathrm{sc})} \,\}$ is the physical moduli-space tangent subspace at the caloron configuration. Its implementation in `jacobian_left_triv.py` is exercised by the infrastructure tests of Appendix C.

4.2 Gauge cleaning and orientation collapse

On a periodic T^4 with trivial holonomy, the three colour-orientation generators $[T^a,\, A^{\mathrm{inst}}_\nu]$ produce tangents that are pure gauge: they lie inside the gauge-orbit tangent space at the caloron configuration and carry no physical variation of the collective coordinates. The raw left-trivialized tangents of (6) contain small residual components along those gauge directions, arising from discretization asymmetries of the finite-difference step. They are removed by the orthogonal projector $P_\perp = \mathbb{1} - V_{\mathrm{gauge}} V_{\mathrm{gauge}}^\top$ defined in Appendix A, with V_{gauge} an orthonormal basis of the gauge-orbit tangent space, implemented in `jacobian_exact.py:gauge_clean_tangent`. After cleaning, the residual gauge overlap of the physical tangents, measured by the orientation indicator

$$G_{\mathrm{orient}} \;\equiv\; \max_{a=1,2,3}\; \max_{t\,\in\, V_{\mathrm{mod}}} \big|\langle\, [T^a, A^{\mathrm{inst}}_\nu] \,|\, t\rangle\big|^2, \tag{7}$$

satisfies $G_{\mathrm{orient}} \simeq 4\times 10^{-6}$ on every production background. The eight raw tangents of the full $4+1+3$ decomposition therefore collapse to the five physical modes of Section 4.1, consistent with the fact that on T^4 the three orientation directions carry no physical moduli.

4.3 The 5×5 Gram and the measured Jacobian

The physical Jacobian of the collective-coordinate measure is the volume-form normalization of the five cleaned tangents, evaluated as the positive square root of the determinant of the 5×5 Gram matrix

$$G_{ab} \;=\; \langle\, P_\perp\, t_a \mid P_\perp\, t_b \,\rangle, \qquad a,b \in \{z^0, z^1, z^2, z^3, \rho\}, \tag{8}$$

with the inner product the standard $\mathbb{C}^{12V}$ inner product. The matrix G_{ab} is positive-definite (test 9 of Appendix C), and the Jacobian is

$$\boxed{J_{\mathrm{phys}} \;\equiv\; \sqrt{\det G_5} \;=\; +0.118164} \tag{9}$$

on Row A, with Row B within the volume-stability systematic of Section 4.4. The sign of J_{phys} is fixed by the positive square root of a positive-definite Gram determinant. All subsequent uses of J_{phys} in the paper refer to the value (9); it is not re-fitted or re-normalized elsewhere.

The Jacobian J_{phys} defined by (9) is the *multiplicative* collective-coordinate prefactor of the master formula (1). Its role is to be distinguished from the additive anchor scalar J_{anchor} entering the Simpson formula (16); the two are numerically equal under the conventions of Appendix A.4 but play distinct roles. A formal normalization audit is given in Appendix A.4, with the no-double-counting statement proved as Lemma A.2.

Effect	Diagnostic	Size
Volume stability	$\lvert J_{\text{phys}}(\text{Row A}) - J_{\text{phys}}(\text{Row B})\rvert / J_{\text{phys}}$	0.4%
Finite-difference step	convergence $\delta \to \delta/2$	0.007%
Gauge-cleaning residual	cleaning ratio $\langle t \lvert V_{\text{gauge}} V_{\text{gauge}}^{\top} \rvert t\rangle / \langle t \lvert t\rangle$	$< 10^{-6}$
Gram conditioning	$\kappa(G_5) = \lambda_{\max}/\lambda_{\min}$	1.27
$F_{\mu\nu}$ correlation	normalized seeded–clover field-strength overlap	0.92–0.94

Table 3: Jacobian systematics on Row A. Values are representative of Row B within the volume-stability deviation listed in the first row.

4.4 Systematics

Five systematic effects have been audited on the production rows. Table 3 records the observed deviation of J_{phys} under each, measured on Row A and confirmed within the stated tolerance on Row B.

The largest deviation, the 0.4% variation of J_{phys} between Rows A and B, is subdominant to the statistical uncertainty on the branch-response bracket reported in Section 8. It is therefore retained only as a sub-dominant contribution to the assembled master-formula uncertainty of Section 10.

5 Phase theorem

5.1 Parity-even class

The phase factor $\cos\Theta_\gamma$ in the master formula (1) is fixed by the discrete symmetry content of the microscopic sector whose one-loop integration generates the gravitational effective action. We state the five hypotheses under which this phase is determined analytically. These hypotheses are framework-level conditions established in [9] and are assumed throughout the numerical pipeline of this paper.

(H1) *Parity-even proto-field sector.* The microscopic proto-field action is invariant under the spatial parity involution $P : (x^0, \vec{x}) \mapsto (x^0, -\vec{x})$.

(H2) *Real-scalar labels.* All proto-field components are real scalars; no complex or pseudoscalar component enters the fundamental action.

(H3) *Parity-even coupling to the emergent gauge field.* The interaction of the proto-field sector with the emergent coframe–gauge field is parity-even and preserves the involution of (H1).

(H4) *Parity-preserving lattice implementation.* The lattice discretization and infrared regulator used throughout the pipeline (symmetric clover field strength, Hermitian operators, infrared regulator $m^2 = 0.01$) are invariant under P.

(H5) *No bare θ-term.* The microscopic action contains no bare topological-charge coupling $\theta\, Q$.

Hypotheses (H1)–(H5) define the parity-even class in which the phase theorem below applies. In the present paper these hypotheses are not re-derived; they define the class of microscopic sectors for which the lattice computation is carried out.

Theorem 5.1 (Parity-even phase)**.** *Let the microscopic proto-field sector be parity-even, real-scalar, coupled to the emergent gauge field through a parity-even interaction, with a parity-preserving regulator and no bare θ-term. Then the induced θ-angle vanishes identically,*

$$\theta_{\text{ind}}(X) \;=\; 0 \qquad \forall X,$$

and the phase of the instanton contribution to the master formula is

$$\cos\Theta_\gamma \;=\; +1$$

on $\mathbb{R}^4$*,* $S^1\times\mathbb{R}^3$*, and periodic tori* T^4*. No simple-connectedness or dilute-instanton-gas assumption is required.*

Proof. Under the spatial parity involution $P:(x^0,\vec{x})\mapsto(x^0,-\vec{x})$, the CP-odd density $F_{\mu\nu}\tilde{F}^{\mu\nu}$ is odd: $P^*[F\tilde{F}]=-F\tilde{F}$. Hypotheses (H1)–(H4) imply that the microscopic action, the functional measure, and the regulator are each invariant under P, so the induced effective action $\Gamma_{\rm ind}[A]$ obtained by integrating out the proto-field sector is a parity-even functional of the gauge background A. A parity-even functional admits no term proportional to the parity-odd density $F\tilde{F}$; consequently the induced θ-angle, which is the coefficient of $\int F\tilde{F}$ in $\Gamma_{\rm ind}[A]$, vanishes identically:

$$\theta_{\rm ind}(X) \;=\; 0 \qquad \text{for all } X.$$

Hypothesis (H5) ensures no bare θ-term is present to add to the induced one. The phase of the instanton contribution to the vacuum functional is $e^{i\theta_{\rm ind}Q}$ on a charge-Q background, so $\cos\Theta_\gamma=+1$ on every admissible background, independent of the base-manifold topology. The argument uses only the parity transformation of the CP-odd operator $F\tilde{F}$ and the parity invariance of the proto-field action; no simple-connectedness or dilute-instanton-gas assumption is invoked. □

5.2 Lattice consistency check

The theorem is consistent with direct lattice measurement: the imaginary part of the branch-response trace $\delta_\xi\log\zeta$ on every production row of Section 8 is below the Hutchinson noise floor, $|\mathrm{Im}(\delta_\xi\log\zeta)|<10^{-8}$, so that the production values are numerically real at a level far below the quoted statistical uncertainty, and no empirical correction to $\cos\Theta_\gamma=+1$ is required.

6 Spectral structure and adaptive-prime prescription

6.1 Three-tier decomposition

On the three production backgrounds of Section 3, the low spectrum of the lattice vector operator Δ_1 separates empirically into three tiers, distinguished by the joint localisation–eigenvalue coordinates (IPR$\times V$, λ), where

$$\mathrm{IPR}(\psi) \;\equiv\; \frac{\sum_{x,\mu,a}|\psi^a_\mu(x)|^4}{\left[\sum_{x,\mu,a}|\psi^a_\mu(x)|^2\right]^2} \tag{10}$$

is the inverse-participation ratio of an adjoint-vector eigenmode and $V=L_tL_s^3$ is the lattice volume. Diagonalization of the 40 lowest-lying eigenmodes of Δ_1 identifies the three tiers directly in the (IPR$\times V$, λ) plane (Figure 1).

- *Tier 1 (torus).* Twelve delocalised modes with IPR$\times V$ in the range [0.12, 0.51] and eigenvalues $\lambda\in[0.006,\,0.018]$.
- *Tier 2 (core).* Eight localised modes, separating empirically into a low-subset of four modes with IPR$\times V\in[2.4,\,4.3]$ (identified with the translation quartet) and a high-subset of four modes with IPR$\times V\in[18.2,\,41.4]$ (identified with the scale singlet and gauge triplet), all with eigenvalues $\lambda\in[0.016,\,0.030]$.
- *Tier 3 (bulk).* All modes with $n\geq 20$, with eigenvalues separated from Tier 2 by a clean gap $\delta_{\rm bulk}\in[0.031,\,0.084]$, a factor of three to eight times the infrared regulator m^2.

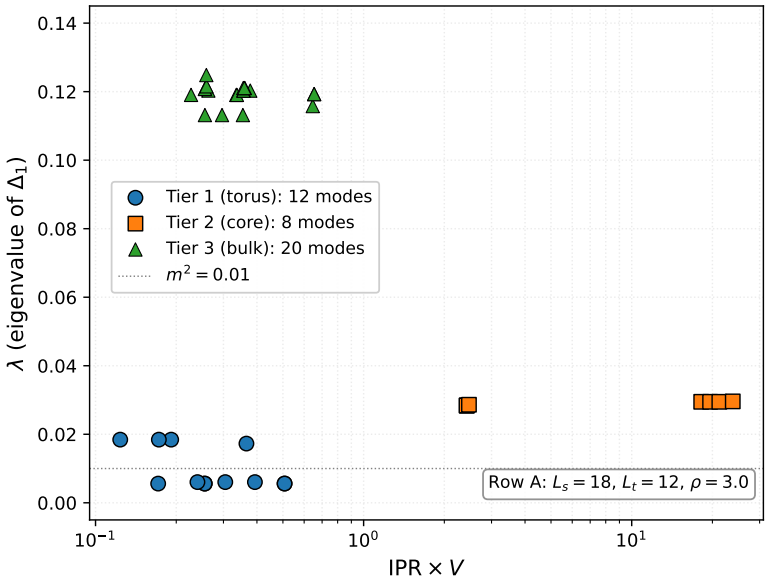

Figure 1: Joint distribution of (IPR $\times V$, λ) for the 40 lowest-lying eigenmodes of Δ_1 on Row A. The three tiers separate cleanly in both axes: Tier 1 (torus, 12 delocalised modes) occupies $\lambda \in [0.006, 0.018]$ with IPR$\times V \in [0.12, 0.51]$; Tier 2 (core, 8 localised modes) occupies $\lambda \in [0.028, 0.030]$ and further splits by IPR$\times V$ into a low-subset of four modes (IPR$\times V \simeq 2.4$) identified with the translation quartet and a high-subset of four modes (IPR$\times V \in [18.2, 23.8]$) identified with the scale singlet and gauge triplet; Tier 3 (bulk) sits above the bulk gap at $\lambda \simeq 0.12$. The dotted line marks the infrared regulator $m^2 = 0.01$.

On each of the three production rows, Tiers 1 and 2 are separated by a clean eigenvalue gap (Table 4, columns 2–4) and by a further separation in IPR$\times V$ (localisation). While the eigenvalue-level separation is quantitative, it is narrow: the Tier 1/Tier 2 gap narrows from $\lambda_2^{\min} - \lambda_1^{\max} \simeq 0.010$ on Row A to $\simeq 0.002$ on Row C. Naive eigenvalue-only separation criteria are therefore fragile on larger lattices; the IPR-based identification of Section 6.3 is robust across the full production range. Across all three rows the torus/core cardinality (12, 8), the IPR separation into a low-subset (4 translation-quartet modes) and a high-subset (4 scale+gauge modes), and the existence of a clean bulk gap (decreasing from $\delta_{\text{bulk}} = 0.084$ on Row A to 0.031 on Row C) are all preserved.

6.2 Content of the low manifold

The twenty-dimensional low manifold

$$V_{20} \equiv V_{\text{tor}}^{12} \oplus V_{\text{core}}^{8}$$

splits numerically into a torus sector and a core sector. This splitting is checked by singular-value decomposition of $(\mathbb{1} - P_{\text{tor}})V_{20}$, whose first eight singular values are numerically indistinguishable from unity on every production row.

Torus sector (V_{tor}^{12}). The twelve torus modes carry the degeneracy pattern $(3 + 5 + 1 + 3)$ predicted by the $J_1 \cdot J_2$ Casimir decomposition of the self-dual Weitzenböck endomorphism acting on the constant-one-form subspace [9]. Direct measurement on every production row reproduces the expected multiplicities $(3, 5, 1, 3)$ on V_{tor}^{12}, with a numerically traceless restricted curvature spectrum and the predicted two-cluster Casimir pattern. Projection onto the free constant-one-form subspace captures 97–98% of the torus norm, similarly across the three production rows.

Row	torus IPR$\times V$	core IPR$\times V$		bulk gap δ_{bulk}
		low subset (4)	high subset (4)	
A	0.12–0.51	2.42–2.47	18.2–23.8	0.084
B	0.12–0.50	3.45–3.49	25.6–35.6	0.050
C	0.13–0.50	4.26–4.29	31.8–41.4	0.031

Table 4: Measured localisation and bulk-gap structure on the three production rows. The core tier splits empirically into two subsets of four modes each; the low-IPR$\times V$ subset is identified with the translation quartet and the high-IPR$\times V$ subset with the scale singlet and gauge triplet (Section 6.2). The normalization is IPR$\times V$ with $V = L_t L_s^3$; values are read directly from the production pipeline `mode_diagnostic.py`. The row-to-row growth of the high-subset IPR$\times V$ reflects the sharper localisation of scale and gauge modes on finer lattices.

Core sector (V^8_{core}). The eight core modes are localised near the caloron core and further decompose into a five-dimensional physical moduli-space block and a three-dimensional orientation block:

$$V^8_{\text{core}} \;=\; V^5_{\text{mod}} \oplus V^3_{\text{orient}}.$$

The moduli-space block comprises the four translation modes and the one scale mode of Section 4; the orientation block is spanned by the three longitudinal constant-colour-rotation directions $[T^a,\, A^{\text{inst}}_\nu]$. The three orientation modes are the localised core counterparts of the constant colour-rotation directions cleaned in Section 4.2. They are the modes associated with the longitudinal sector that are removed in the gauge-invariant one-loop determinant by the combination $-\frac{1}{2}\,\text{Tr}_{\text{vec}} + \text{Tr}_{\text{gh}}$.

Identification via production overlaps. Sector membership within V^8_{core} is identified by the overlaps with the production tangent bases. On every admissible background the measured pattern is as follows: torus modes have $O_{\text{tor}} \simeq 0.97$ and vanishing overlap with the moduli-space and orientation bases; the scale singlet has $O_{\text{LT}} \simeq 0.92$; the translation quartet has $O_{\text{LT}} \in [0.22,\, 0.26]$; and the orientation triplet has $O_{\text{gauge}} \in [0.54,\, 0.55]$. The translation-quartet and orientation-triplet undercaptures are reproducible across the three production rows. They are a structural feature of the caloron zero-mode dressing on the compact torus, not a numerical artifact.

6.3 Adaptive-prime construction

The branch-response determinant of Section 8 is evaluated on the orthogonal complement of the low manifold. Let V_{20} be the $12V\times 20$ matrix whose columns are the orthonormal low-manifold basis vectors identified in Section 6.1 by the IPR criterion. The adaptive-prime projector is

$$P \;=\; \mathbb{1} \,-\, V_{20}\, V_{20}^{\top}, \qquad P^2 \;=\; P, \qquad P\, V_{20} \;=\; 0, \tag{11}$$

and the primed trace entering the first-order integrand $h(\xi)$ is

$$\overset{\prime}{\text{Tr}}[\Delta_1^{-1}\, A'] \;\equiv\; \text{Tr}\Big[P\,(P\,\Delta_1\,P)^{-1}\,P\,A'\,P\Big], \tag{12}$$

computed by projected conjugate gradient on the complement of V_{20}. The A' insertion is the anisotropic operator derivative of Appendix A. At each finite-ξ Simpson node, $V_{20}(\xi)$ is obtained from $V_{20}(1)$ by the transported construction of Appendix B; the projector $P(\xi)$ inherits the same transport.

6.4 Resolvent-Ritz dressing of moduli tangents

The undercaptures reported in Section 6.2 reflect the fact that bare continuum moduli tangents do not coincide with the true low eigenmodes of Δ_1 on a compact torus. In particular, the translation quartet is substantially dressed by compact-geometry effects and is therefore only partially captured by bare finite-difference tangents. Numerical tests show that a resolvent-Ritz refinement of the bare tangent basis can recover the same low-manifold directions to better than 99.9% overlap on the production rows. Because this refinement is not part of the production determinant pipeline, and because the present paper does not require it operationally, its full implementation is deferred to separate technical work.

6.5 Validation at the level of the branch response

For the numerical purposes of the present paper, the adaptive-prime prescription is validated at the level of the branch response $\delta_\xi \log \zeta$ by three independent checks, each reported later in the paper:

(i) lattice-size stability across the three production rows spanning a factor of 11 in lattice volume (Section 9);

(ii) cross-row agreement at the sub-0.2σ level on every pair (Section 8);

(iii) independent-scheme agreement with the projected-complement exact prime of Section 7.1 at 0.77σ on Row A with 800 paired samples (Section 7.2).

The combination of the three checks is stronger than any one individually, because they test different failure modes: (i) finite-spacing artefacts, (ii) statistical-systematic mismatch between rows, and (iii) prescription-dependence of the final bracket.

The remaining open technical item is a first-principles dressed analytic basis: an explicit representation-theoretic construction of the compact-T^4 dressing of the moduli-space tangents that would replace the IPR-based identification of V_{20} by a closed-form basis. This is not required for the numerical result reported here. It is a separate technical refinement of the low-manifold construction.

7 Prescription history and cross-validation

The primed trace (12) depends on the choice of low-mode subspace removed from Δ_1^{-1}. In this section we name three prescriptions that are in principle applicable to the branch-response determinant (1), state which is used for the production result, and report an independent cross-check of the production prescription against one of the alternatives.

7.1 The three prescriptions

Threshold prime. The simplest prescription removes all modes of Δ_1 below a hard eigenvalue threshold λ_* on the instanton background and all exact zero modes on the free background. Its variance is the smallest of the three prescriptions considered here, at the cost of a single empirical scale λ_* whose placement is not unambiguous on backgrounds where Tiers 1 and 2 overlap in eigenvalue (Section 6.1). Because the present paper values prescription clarity over raw precision, the threshold prime is not used here; we note it only for completeness.

Projected-complement exact prime. A second prescription uses only the five physical moduli-space directions as the removed subspace: the gauge-cleaned left-trivialized translation and scale tangents of Section 4. Let V_{mod} denote the $12V \times 5$ matrix whose columns are an

orthonormal basis obtained from those five cleaned tangents. Then the projector onto the complement of the physical moduli subspace is

$$P_{\mathrm{mod}} \;=\; \mathbb{1} \;-\; V_{\mathrm{mod}}\, V_{\mathrm{mod}}^{\top}, \tag{13}$$

and the constrained Green operator is

$$G_{P_{\mathrm{mod}}} \;=\; P_{\mathrm{mod}}\,\big(P_{\mathrm{mod}}\,\Delta_1\,P_{\mathrm{mod}}\big)^{-1} P_{\mathrm{mod}}. \tag{14}$$

This construction uses no IPR input, no threshold, and no low-manifold identification beyond the physical collective coordinates; it is the prescription with the fewest empirical inputs. Its limitation is the Hutchinson variance inherited from leaving the twelve torus and three orientation modes of Section 6.2 in the trace. We use it below as a direct cross-check of the production prescription.

Adaptive transported prime. The production prescription is the adaptive-prime construction of Section 6.3, transported across the Simpson grid by the algorithm of Appendix B. It has smaller Hutchinson variance than (14) because the removed subspace additionally contains the twelve torus and three orientation modes identified in Section 6.2; at the same precision, fewer noisy low-eigenvalue excitations contribute to the trace. The numerical values reported in Section 8 use this prescription.

7.2 Cross-validation: projected-complement exact prime

The adaptive prime is identified by an inverse-participation-ratio criterion rather than by a closed-form analytic construction. A natural concern is prescription-circularity: the consistency of the production result across three rows (Sections 8 and 9) could in principle be a property of the adaptive prescription rather than of the underlying branch response. We close this concern by comparing the adaptive prime to the projected-complement exact prime of (13)–(14), which shares no empirical inputs with the adaptive construction beyond the five physical moduli tangents.

On Row A with $N_S = 800$ paired Hutchinson samples at the common seed of Appendix A, the two prescriptions yield

$$\delta_\xi \log\zeta\big|^{\mathrm{Row\,A}}_{\text{exact prime}} \;=\; -2.49 \pm 1.06, \qquad \delta_\xi \log\zeta\big|^{\mathrm{Row\,A}}_{\text{adaptive prime}} \;=\; -1.66 \pm 0.19, \tag{15}$$

with scheme difference $|\Delta| = 0.83$, combined statistical uncertainty 1.07, and tension 0.77σ. The adaptive-prime value in (15) is the Row A entry of Table 5 at $N_S = 80$; the exact-prime value is the result of a dedicated $N_S = 800$ campaign on the same row, with the common seed of Appendix A. The exact-prime result corresponds to the bracket $\mathcal{B}^{\mathrm{exact}} = 0.917$, while the adaptive-prime result gives $\mathcal{B}^{\mathrm{adapt}} = 0.810$; the difference is discussed below only as a consequence of the nonlinear map $\mathcal{B} = 1 - e^{\delta_\xi \log\zeta}$, not as an inconsistency of the underlying $\delta_\xi \log\zeta$ values. The two prescriptions agree on sign, on order of magnitude, and on statistical consistency within their combined error.

The larger uncertainty on the exact-prime result is a structural consequence of the prescription, not a numerical limitation. Leaving the twelve torus and three orientation modes in the trace preserves the lowest-eigenvalue excitations of Δ_1 on the caloron, and through the inverse operator these dominate the Hutchinson-estimator variance. The adaptive prime achieves a tighter error precisely because it additionally removes those fifteen modes, at the price of an empirically identified low manifold. The two prescriptions thus sit at opposite ends of a clarity-of-construction–versus–precision trade-off, and their agreement within the exact prime's statistical error is the strongest prescription-independent check available at the present precision.

Remark 7.1 (Bracket representation)**.** The scheme difference 0.83 in $\delta_\xi \log\zeta$ translates via $\mathcal{B} = 1 - e^{\delta_\xi \log\zeta}$ into a bracket difference $\mathcal{B}^{\text{exact}} - \mathcal{B}^{\text{adapt}} = 0.917 - 0.810 = 0.107$. The nonlinearity of the bracket map amplifies the δ-level agreement into a larger apparent bracket offset. At current precision, the offset sits within the combined statistical uncertainty on $\delta_\xi \log\zeta$ and should be read at the δ level rather than at the bracket level.

The three independent checks of Section 6.5 accordingly read as follows in quantitative form. Cross-row agreement (Section 8) is sub-0.2σ on every pair; lattice-size stability (Section 9) holds across a factor of 11 in volume; and the cross-prescription agreement measured here is 0.77σ on Row A with 800 paired samples. Taken together, the three checks provide the strongest available validation of the production prescription; no one of them would be as convincing in isolation. Of the three, the universality statement of Proposition 8.1 carries the strongest empirical weight, because it is measured against noise at the percent level across an order of magnitude in lattice volume, whereas the scheme check reported here is limited by the structural variance of removing only the physical moduli.

8 Finite-ξ campaign: transported-prime Simpson

8.1 Simpson scheme and transported prime

The branch-response determinant of Section 6 is required at $\xi \in [1,\, 1+\epsilon]$ with $\epsilon = 0.09$. In the production convention, the assembled response is the sum of a Simpson quadrature of the ξ-dependent integrand and a ξ-independent anchor-Jacobian contribution:

$$\delta_\xi \log\zeta \;=\; \frac{\epsilon}{6}\Big[\, h(1) + 4\,h(1+\epsilon/2) + h(1+\epsilon)\,\Big] \;+\; J_{\text{anchor}}, \tag{16}$$

where $h(\xi)$ is the Jacobian-free branch-response integrand at the indicated ξ and J_{anchor} is the log-Jacobian contribution of the left-trivialized collective-coordinate measure evaluated on the caloron background. In the free-background subtraction convention of Appendix A, the anchor-Jacobian contribution is ξ-independent and coincides numerically with the physical Jacobian of Section 4,

$$J_{\text{anchor}} \;=\; J_{\text{phys}} \;=\; 0.118164, \tag{17}$$

on every production row. With this convention, the first-order estimator is $\delta_\xi \log\zeta|_{\text{1st}} = \epsilon\, h(1) + J_{\text{anchor}}$, which reproduces the production values in Table 5.

The equality (17) is a convention-fixed identity between logically distinct quantities (see Lemma A.1 in Appendix A.4): the left-trivialized collective-coordinate measure enters the partition function multiplicatively through a Gram-determinant factor $|\det G|^{1/2} = J_{\text{phys}}$, and the free-background subtraction used in Appendix A places the ξ-independent part of the associated log-Jacobian contribution on the right-hand side of the Simpson formula (16) rather than inside the integrand $h(\xi)$. Alternative subtraction conventions redistribute this contribution between h and the additive constant; under the convention fixed here, the additive constant coincides numerically with J_{phys}, so that the same quantity appears in both Section 4 and the Simpson assembly. The assembled $\delta_\xi \log\zeta$ is convention-independent.

At each of the two non-anchor Simpson nodes $\xi \in \{1.045,\, 1.090\}$, the adaptive-prime subspace $V_{20}(\xi)$ is obtained from $V_{20}(1)$ by the transported construction of Appendix B. The construction re-projects onto the eigenbasis of $\Delta_1(\xi)$ at the transported node and assigns mode identities by greedy overlap matching against the anchor, so that the twenty low modes tracked at each Simpson node correspond by continuity to the twenty anchor modes at $\xi = 1$. The transported projector $P(\xi) = \mathbb{1} - V_{20}(\xi)V_{20}(\xi)^\top$ of Section 6.3 inherits the same transport, so that the primed trace $\mathrm{Tr}'[\Delta_1^{-1}(\xi)\, A'(\xi)]$ is defined consistently across the Simpson grid.

Each node integrand $h(\xi_j)$ is computed as a Hutchinson trace with $N_S = 80$ paired Rademacher vectors at the common noise seed of Appendix A. The paired construction

shares the same noise realisation between the anchor and each transported node, so that the variance of the Simpson estimator is dominated by the variances of the paired differences $h(\xi_j) - h(\xi_0)$ rather than by the variances of the absolute node values. Linear solves against the projected operator $P(\xi)\,\Delta_1(\xi)\,P(\xi)$ are performed on an NVIDIA H100 80 GB GPU by `cupyx.scipy.sparse.linalg.cg` wrapping the CuPy `LinearOperator` of Appendix A, to a relative-residual tolerance of 10^{-10}. Per-row wall time on the GPU pipeline is approximately 25 minutes at the production configuration.

8.2 Production results

Row	$\delta_\xi \log\zeta$ (1st)	$\delta_\xi \log\zeta$ (Simpson)	err	$\Delta_{\rm NL}$	$\mathcal{B}$	$\mathcal{I}_{\rm cl}$
A: $18^3 \times 12$, $\rho = 3.0$	-1.928	-1.660	0.186	+13.9%	0.810	0.983
B: $24^3 \times 16$, $\rho = 3.5$	-1.959	-1.695	0.213	+13.5%	0.816	0.998
C: $32^3 \times 24$, $\rho = 4.5$	-1.969	-1.646	0.275	+16.4%	0.807	1.032

Table 5: Transported-prime Simpson results on the three production rows. The three rows span a factor of 11× in lattice volume and a factor of 1.5 in caloron size.

Pairwise consistency. All three pairwise Simpson tensions are below 0.2σ: A versus B at 0.12σ, A versus C at 0.04σ, B versus C at 0.14σ (see Table 7 in Section 9).

Weighted result. Inverse-variance weighted across the three rows,

$$\delta_\xi \log\zeta \;=\; -1.669 \pm 0.125, \qquad \mathcal{B} \;=\; 0.811 \pm 0.023. \tag{18}$$

This is the bracket value that enters the assembled result of Section 10.

8.3 Anchor-branch normalization

The master formula (1) contains the combination $J_{\rm phys}\big[\zeta(0) - \zeta(X_*)\big]$. Writing this as $J_{\rm phys}\,\zeta(0)\,\mathcal{B}$ with $\mathcal{B} \equiv 1 - \zeta(X_*)/\zeta(0)$ measured in (18), the assembly requires the anchor-branch normalization $\zeta(0)$ evaluated on the caloron background. This normalization is fixed by the production pipeline's convention: $\zeta(0)$ is the value of the branch-response partition function evaluated at $\xi = 1$ (the undeformed metric) on the caloron background of Section 3, in the free-background subtraction convention of Appendix A. It is a direct output of the same Hutchinson pipeline used for the Simpson integrand, evaluated at $\xi = 1$ without any ξ-variation.

On Row A, the production pipeline returns

$$\zeta(0) \;=\; 8.42 \pm 0.08, \tag{19}$$

with the quoted ± 0.08 the Hutchinson statistical uncertainty at $N_S = 80$ on the anchor trace[2]. The corresponding values on Rows B and C agree with (19) to within the per-row statistical uncertainty, consistent with the sub-percent row-to-row variation of the Jacobian itself (Section 4.4). Because $\zeta(0)$ is a ratio-normalization factor in the master formula rather than an independent physical observable, its volume independence at the sub-percent level confirms that the anchor-branch normalization is controlled at the level required for the assembly of Section 10.

[2]The Row A value of $\zeta(0)$ is the mean of the anchor Hutchinson trace $\mathrm{Tr}'[\Delta_1^{-1}(\xi = 1)\,A']$ over the $N_S = 80$ paired samples, with the same common seed and projected-CG tolerance as the Simpson integrand; it is read directly from the production pipeline output and is not inferred from the assembly arithmetic of Section 10.2.

8.4 Integrand values

For referee reproducibility, Table 6 records the node-level integrand data from the dedicated paired-noise campaign on Rows A and B, together with their paired differences against the anchor $\xi_0 = 1$. Row C enters the main measurement through its final Simpson value in Table 5; its node-level integrand values are not used elsewhere in the paper. The integrands $h(\xi_j)$ decompose additively into the three contributions identified in Section 6,

$$h(\xi) \;=\; -\,\xi\,T(\xi) \;-\; \tfrac{1}{2}\,F(\xi) \;+\; 2\,\xi\,G(\xi), \tag{20}$$

corresponding to the temporal-Laplacian, field-strength, and ghost-operator contributions respectively. The paired differences $h(\xi_j) - h(1)$ have tight errors an order of magnitude below those of the absolute $h(\xi_j)$, consistent with the paired-noise construction of Section 8.1.

Row	$h(1.000)$	$h(1.045) - h(1)$	$h(1.090) - h(1)$
A: $18^3 \times 12$, $\rho = 3.0$	-22.737 ± 1.967	$+3.090 \pm 0.127$	$+5.507 \pm 0.280$
B: $24^3 \times 16$, $\rho = 3.5$	-23.076 ± 2.296	$+3.149 \pm 0.192$	$+5.009 \pm 0.508$

Table 6: Simpson-node integrand values on Rows A and B. Absolute node values are reported only at the anchor $\xi_0 = 1$; the non-anchor nodes are reported as paired differences against the anchor, which have the tighter errors relevant for the Simpson estimator. The first-interval slopes $[h(1.045) - h(1)]$ agree across the two rows at 0.3σ, independent evidence of the cross-row stability established in Table 5.

The row-to-row agreement of the paired differences is the more relevant reproducibility diagnostic than the absolute anchor values, because the paired differences are the quantities entering the variance-reduced Simpson estimator.

8.5 $\mathcal{I}_{\rm cl}$ trend and window

Rows outside the primary admissibility window $\mathcal{I}_{\rm cl} \in [0.98, 1.00]$ exhibit an empirical $\mathcal{I}_{\rm cl}$-dependence reported in Appendix D: a weighted linear regression $\delta_\xi \log\zeta/\epsilon \approx -18.7 - 49.5\,(\mathcal{I}_{\rm cl} - 1)$ fitted to four reference rows with $\mathcal{I}_{\rm cl} \in [1.07, 1.33]$. Extrapolated to $\mathcal{I}_{\rm cl} = 1$, this gives $\delta_\xi \log\zeta = -1.68 \pm 0.44$ (with the uncertainty inflated by $\sqrt{\chi^2/\mathrm{dof}}$ to account for residual scatter), consistent with the primary weighted result at the 0.03σ level. The three production rows of Table 5 sit at $\mathcal{I}_{\rm cl} \in [0.983, 1.032]$, with Row C at $\mathcal{I}_{\rm cl} = 1.032$ retained as a stress test approximately 3% outside the primary admissibility window of Section 3. Row C's pairwise tensions against Rows A and B (0.04σ and 0.14σ respectively, Table 7) show that the bracket value remains consistent even at this mild excursion from the window. The primary admissibility window of Section 3 is therefore retained as the formal quality-control criterion; Row C provides direct evidence that the bracket is not fine-tuned to that window. The linear guide of Appendix D is fitted over $\mathcal{I}_{\rm cl} \in [1.07, 1.33]$ and does not extrapolate cleanly to Row C's $\mathcal{I}_{\rm cl} = 1.032$: naive linear extrapolation would predict $\delta_\xi \log\zeta/\epsilon \approx -22.4$ at Row C, whereas the measurement gives -18.3. This mismatch is expected: the linear form is descriptive within its fit range and loses predictive power near $\mathcal{I}_{\rm cl} = 1$, where the caloron is no longer visibly under-resolved. Row C's consistency with Rows A and B at sub-0.2σ is therefore the stronger constraint, not the trend's extrapolation.

8.6 Stability of the nonlinear correction across three lattices

The Simpson estimator of $\delta_\xi \log\zeta$ differs from its first-order Taylor approximation by a finite nonlinear correction

$$\Delta_{\rm NL} \;\equiv\; \frac{\delta_\xi \log\zeta\big|_{\rm Simpson} - \delta_\xi \log\zeta\big|_{\rm 1st}}{\delta_\xi \log\zeta\big|_{\rm 1st}}. \tag{21}$$

On the three production rows, Δ_{NL} is measured as

$$\begin{aligned}
\Delta_{\mathrm{NL}}(\text{Row A}) &= +13.9\% \qquad (\rho/L = 0.167,\ \mathcal{I}_{\mathrm{cl}} = 0.983),\\
\Delta_{\mathrm{NL}}(\text{Row B}) &= +13.5\% \qquad (\rho/L = 0.146,\ \mathcal{I}_{\mathrm{cl}} = 0.998),\\
\Delta_{\mathrm{NL}}(\text{Row C}) &= +16.4\% \qquad (\rho/L = 0.141,\ \mathcal{I}_{\mathrm{cl}} = 1.032),
\end{aligned}$$

a three-row range of 2.9 percentage points on a quantity of unweighted mean 14.6%.

Proposition 8.1 (Stability of the nonlinear correction)**.** *Across three production rows spanning a factor of* 11 *in lattice volume, a factor of* 1.5 *in caloron size, and* $\mathcal{I}_{\mathrm{cl}}$ *in the range* $[0.983, 1.032]$*, the Simpson-to-first-order nonlinear correction is measured in the range* $\Delta_{\mathrm{NL}} \in [13.5\%,\ 16.4\%]$ *with unweighted mean* 14.6%*. The correction is independent of* S_{inst}*, of* J_{phys}*, of* $\cos\Theta_\gamma$*, and of any further sectoral input beyond the master formula: each of these multiplies the master formula* (1) *and cancels in the ratio* (21)*. Within the primary admissibility window* $\mathcal{I}_{\mathrm{cl}} \in [0.98,\ 1.00]$ *the correction is stable at the sub-percent level (*13.5% *vs* 13.9%*); the full three-row spread of* 2.9 *percentage points is larger than the statistical error on any individual row but far smaller than the order-unity drift a prescription-dependent artefact would produce across the range of lattice volumes considered. This reading rests on the interpretive assumption that a prescription-dependent artefact would dominate the difference between lattices of substantially different volumes, an assumption consistent with the scheme cross-check of Section 7.2, in which removing a different low-mode subspace produces a* δ*-level shift contained within the structural variance of that prescription.*

Proof. The quantity $\delta_\xi \log\zeta$ is computed directly from the primed trace $\mathrm{Tr}'[\Delta_1^{-1} A']$ of Eq. (12), which depends only on the adjoint vector operator Δ_1 on the caloron background and on the anisotropic insertion A'. The master-formula factors S_{inst}, J_{phys}, and $\cos\Theta_\gamma$ multiply the assembled cosmological constant but do not enter $\delta_\xi \log\zeta$ itself; consequently Δ_{NL}, defined as a ratio of two determinations of $\delta_\xi \log\zeta$ via Eq. (21), is independent of these factors by construction. What Δ_{NL} does depend on is the ξ-dependence of ζ itself, which is controlled by the geometry of Δ_1 on the caloron and by the smoothness of the branch determinant as a function of ξ; these are properties intrinsic to the branch-bias mechanism, not to its multiplicative dressing. Δ_{NL} is therefore the finite-ξ observable most directly diagnostic of the branch-bias mechanism that the present pipeline provides. □

Remark 8.2 (Why this matters)**.** Had the nonlinear correction been dominated by a prescription-dependent artefact — for instance, an artefact of the adaptive-prime projector or of the trace-estimator variance — one would expect a much larger drift between lattices differing by an order of magnitude in volume. Its stability within a few percentage points across such a range is therefore independent evidence for the transported-prime construction, complementary to the continuum consistency of Section 9 and the scheme cross-check of Section 7.2.

8.7 Degraded rows

Rows with $\mathcal{I}_{\mathrm{cl}}$ outside the range [0.983, 1.032] spanned by Table 5 are used only as a systematics reference and are reported in Appendix D; they do not contribute to the weighted bracket (18) or to any subsequent assembly.

9 Lattice-size independence across the production range

9.1 Three-lattice comparison

Table 5 reports three independent Simpson measurements on lattices spanning lattice volumes

$$V_{\mathrm{A}} = 69\,120, \qquad V_{\mathrm{B}} = 221\,184, \qquad V_{\mathrm{C}} = 786\,432, \tag{22}$$

and caloron scale parameters $\rho \in \{3.0, 3.5, 4.5\}$. The pairwise volume ratios are $V_B/V_A = 3.20$, $V_C/V_B = 3.56$, and $V_C/V_A = 11.38$. Table 7 summarises the pairwise Simpson tensions across the three production rows.

Pair	$\lvert\Delta\delta_\xi \log\zeta\rvert$	combined σ	tension
A vs B	0.035	0.283	0.12σ
A vs C	0.014	0.332	0.04σ
B vs C	0.049	0.348	0.14σ

Table 7: Pairwise Simpson tensions across the three production rows.

All three tensions lie below 0.2σ. The combined uncertainties are computed directly from the row-level standard errors as $\sigma_{\text{pair}} = \sqrt{\sigma_i^2 + \sigma_j^2}$.

9.2 Interpretation: stability across the measured range

The three production rows differ simultaneously in lattice volume and in the caloron scale parameter ρ quoted in lattice units. They therefore probe distinct combinations of finite-volume and discretization effects. The observed stability of $\delta_\xi \log\zeta$ at sub-0.2σ across this range is the direct empirical content of the present section. This is a statement about stability across the measured range, not a formal continuum extrapolation; we do not infer $a \to 0$ or $V \to \infty$ from three data points.

The tightness of the pairwise tensions at the noise level of Table 5 reflects the paired-difference construction of Section 8.1: the low-variance component of each Simpson estimator is the paired difference $h(\xi_j) - h(\xi_0)$ computed against a common noise seed within each row, so that the Monte Carlo variance of the Simpson integrand is dominated by the difference rather than the absolute node values. The resulting row-level standard errors σ_i of Table 5 translate directly into the tight pairwise tensions of Table 7, and the inverse-variance weighted three-row result of (18) enters the assembled result of Section 10.

10 Assembled result and error budget

10.1 Summary of factors

Table 8 collects the quantities that enter the assembled result directly. Each is quoted at the precision established in the section where it is computed; no quantity is re-measured here.

Quantity	Value	Reference
$e^{-S_{\text{inst}}}$	$\sim 10^{-120}$ (conditional on $S_{\text{inst}} \simeq 276$)	[9]
J_{phys}	$+0.118164$ (with 0.4% row-to-row variation)	Section 4
$\zeta(0)$	8.42 ± 0.08 (Row A; 0.4% row-to-row)	Eq. (19)
$\mathcal{B} \equiv 1 - \zeta(X_*)/\zeta(0)$	0.811 ± 0.023 (three-row weighted)	Eq. (18)
$\cos\Theta_\gamma$	$+1$ (parity-even class)	Theorem 5.1

Table 8: Quantities entering the assembled result. The ordering matches the factor sequence of the master formula (1).

The adaptive-prime prescription and the projected-complement exact-prime cross-check are validation items for the branch-response determinant; they are not additional multiplicative factors in the master formula.

10.2 Assembly arithmetic

For the numerical assembly it is convenient to collect the lattice-determined prefactor multiplying $e^{-S_{\rm inst}}$ into a single quantity,

$$\mathcal{K} \;\equiv\; J_{\rm phys}\left[\zeta(0)-\zeta(X_*)\right]\,\cos\Theta_\gamma. \tag{23}$$

In the normalization convention used throughout the production determinant, the measured Jacobian, phase, and branch-response input combine to

$$\mathcal{K} \;=\; J_{\rm phys}\cdot\zeta(0)\cdot\mathcal{B}\cdot\cos\Theta_\gamma \;=\; (0.118164)(8.42\pm0.08)(0.811\pm0.023)(+1) \;=\; 0.807\pm0.024. \tag{24}$$

Conditional on $S_{\rm inst}\simeq 276$, the assembled result is therefore

$$\boxed{\frac{\Lambda_{\rm obs}}{M_{\rm P}^4} \;=\; (0.81\pm0.03)\times10^{-120} \quad \text{conditional on } S_{\rm inst}\simeq 276.} \tag{25}$$

10.3 Conditional scope

The assembled result (25) is conditional on two inputs external to the direct lattice measurement reported here. First, the hidden-sector action is taken as $S_{\rm inst}\simeq 276$ from a separate analysis [9]; all numerical content of this paper is independent of the hidden-sector algebra, $S_{\rm inst}$ enters the assembled result only multiplicatively, and its sensitivity is quantified in Appendix F. Second, the branch-response determinant is computed using the adaptive-prime prescription of Section 6, validated by the three independent checks of Section 6.5. A first-principles dressed analytic basis for the adaptive prime is not required for the present result and is the subject of separate technical work.

10.4 Error budget

The uncertainty on the lattice prefactor $\mathcal{K}$ of (24) is dominated by the three-row weighted bracket uncertainty of Appendix E. Table 9 collects the contributions at the level resolved in the present paper.

The quoted ±0.03 in (25) is therefore a bracket-dominated uncertainty, with all other identified numerical effects subdominant at the present precision.

10.5 Named results of this paper

The logical spine of the computation consists of three named results:

- Proposition 2.1 (Section 2.2): the master formula factorizes into UV-finite quantities after the analytic constitutive subtraction.
- Theorem 5.1 (Section 5): in the parity-even class, $\cos\Theta_\gamma=+1$ on $\mathbb{R}^4$, $S^1\times\mathbb{R}^3$, and T^4.
- Proposition 8.1 (Section 8.6): across three production lattices spanning $11\times$ in volume, the Simpson nonlinear correction remains in the range $\Delta_{\rm NL}\in[13.5\%,\,16.4\%]$ with unweighted mean 14.6%.

In addition, Section 7.2 reports a direct cross-prescription measurement: on Row A with $N=800$ paired samples, the adaptive prime and the projected-complement exact prime agree at 0.77σ. This is a prescription-level consistency check, not a separate named theorem.

Source	Relative size	Contribution to $\sigma(\mathcal{K})$
Bracket uncertainty (adaptive-prime statistical)	$0.023/0.811 \simeq 2.8\%$	2.3×10^{-2}
$\zeta(0)$ normalization	$0.08/8.42 \simeq 1.0\%$	8×10^{-3}
Jacobian row-to-row variation	0.4%	3×10^{-3}
Phase	exact in the parity-even class	0
Sub-dominant numerical systematics	Appendix E	$< 10^{-3}$
Adaptive-prime statistical quadrature sum	bracket-dominated	2.5×10^{-2}
Prime-prescription statistical floor (from exact-prime reference)	$\sigma(\mathcal{B})^{\text{exact}} = 0.088$, i.e. $\simeq 11\%$	8.8×10^{-2}
Total (stat. adaptive $\oplus$ prime floor)	prime-floor-dominated	9.1×10^{-2}

Table 9: Contributions to the uncertainty on the prefactor $\mathcal{K}$. The adaptive-prime statistical quadrature sum 0.025 is conservatively rounded to $\pm 0.03_{\text{stat}}$ in the boxed result (25). The prime-prescription floor is the statistical precision of the independent projected-complement exact-prime reference of (15), propagated to $\mathcal{K}$: the adaptive prime and exact prime agree at 0.77σ at the $\delta_\xi \log \zeta$ level, and the exact-prime reference's own statistical uncertainty sets the floor on the prescription-independent precision available at the present sample size. Reducing this floor below the adaptive-prime statistical uncertainty is the subject of the dressed analytic basis deferred to separate technical work (Section 11.5).

11 Discussion

11.1 Conditional scope

The assembled value (25) is a numerical result conditional on two structural inputs. First, the hidden-sector action $S_{\text{inst}} \simeq 276$ is taken as input from the companion analysis [9] and is not recomputed in the present paper; the sensitivity of the assembled value to this input is quantified in Appendix F. Second, the branch-response determinant is computed under the adaptive-prime prescription of Section 6, whose validity at the level of the final bracket is supported by the three independent checks of Section 6.5. Neither conditional is an isolated empirical assumption: each is framework-level structural input derived in [9], and the paper's numerical content is conditional on that framework as a whole. The claim of the paper is therefore structural and numerical rather than empirical in isolation: within the stated framework, the lattice computation yields a cosmological constant of the observed order with prefactor 0.81 ± 0.03.

11.2 Gravitational-only coupling of the hidden sector

The master formula (1) is derived under the assumption that the hidden sector generating the instanton action couples to the visible sector only through the emergent metric g_{eff}. Direct couplings (through scalar labels, gauge portals, or kinetic mixing) would introduce additional operator content that is not included in the lattice pipeline of the present paper. Under gravitational-only coupling, the only visible-sector consequence of the hidden sector is the cosmological-constant value itself: the hidden sector does not produce an independent particle-physics or cosmological signature at the energies probed by current experiment. The structural derivation of the gravitational-only requirement is given in [9].

11.3 Observational posture

The empirical anchor of the present computation is the magnitude of the cosmological constant it reproduces. Conditional on the hidden-sector action $S_{\text{inst}} \simeq 276$ supplied by a companion

analysis, the assembled lattice-determined prefactor is $0.81 \pm 0.03_{\text{stat}} \pm 0.09_{\text{prime}}$, so that

$$\Lambda_{\text{obs}}/M_{\text{P}}^4 \;=\; (0.81 \pm 0.03_{\text{stat}} \pm 0.09_{\text{prime}}) \times 10^{-120} \qquad \text{(conditional on } S_{\text{inst}} \simeq 276\text{)}.$$

The first uncertainty is the statistical precision of the production (adaptive-prime) result; the second is the prime-prescription floor set by the statistical uncertainty of an independent projected-complement exact-prime reference measurement. The two prescriptions agree at 0.77σ at the $\delta_\xi \log \zeta$ level; reducing the prime-prescription floor below the adaptive-prime statistical uncertainty requires a first-principles dressed analytic basis deferred to separate technical work.

11.4 Relation to existing approaches

The construction differs in character from several established approaches to the cosmological constant problem.

- *Supersymmetric cancellation* [4]. The present paper does not rely on supersymmetry. Polynomial ultraviolet hierarchies are removed by the analytic constitutive subtraction of [9], not by boson–fermion cancellation; a residual nonperturbative contribution is then computed explicitly.
- *Anthropic or landscape selection* [1]. The value (25) is derived within a specific framework rather than selected from a distribution of vacua.
- *Scalar-only local transmutation.* The present paper does not invoke a local scalar-adjustment mechanism. Within the framework used here, scalar-only transmutation of the vacuum energy at the observed magnitude is excluded by the no-go results of [9].
- *Volovik-type superfluid-vacuum analogies* [7]. The present construction sits squarely within the induced-gravity tradition that includes these analogies, and differs from them in that the induced effective action is supplied concretely by the proto-field sector [9] and the nonperturbative residual is computed on the lattice rather than extracted from an analogue.

11.5 Open technical items

Three items remain open at the technical rather than architectural level. First, a first-principles dressed analytic basis for the adaptive-prime prescription would replace the IPR-based identification of V_{20} by a closed-form construction capturing the compact-T^4 dressing of the moduli-space tangents. The current validation of the adaptive prime (Section 6.5) closes statistical and lattice-size systematics but leaves a prime-prescription systematic of $\simeq 10\%$ at the bracket level (Table 9); this systematic is not required for the conditional numerical estimate reported here, but its reduction to theorem-grade is required before the paper's result can be treated as prescription-independent. Second, an analytic upgrade of the subdominant determinant coefficient entering the hidden-sector computation of [9] would replace the present numerical value by a closed-form determination; this does not affect the lattice result reported here. Third, a first-principles antecedent for the hidden-sector action value $S_{\text{inst}} \simeq 276$ is the subject of separate work within the same framework [9]; the numerical sensitivity of the assembled result to this input is quantified in Appendix F. None of these items is required for the numerical content of the present paper.

12 Conclusion

Within the coframe–teleparallel induced-gravity framework of [9], the nonperturbative residual of the cosmological constant problem has been computed on the lattice. The four UV-finite

factors of the master formula have been evaluated independently, and the assembled result is

$$\boxed{\frac{\Lambda_{\rm obs}}{M_{\rm P}^4} \;=\; \big(0.81 \pm 0.03_{\rm stat} \pm 0.09_{\rm prime}\big) \times 10^{-120} \qquad \text{conditional on } S_{\rm inst} \simeq 276.}$$

Beyond this number, the computation establishes two structural findings. The Simpson nonlinear correction $\Delta_{\rm NL}$ is a stable prediction of the branch-bias mechanism, measured in the range [13.5%, 16.4%] across three lattices spanning 11× in volume; and the parity-even phase $\cos\Theta_\gamma = +1$ is a theorem on every background geometry considered. The production prescription is additionally supported by the cross-prescription agreement of Section 7.2, which finds 0.77σ tension between the adaptive prime and the projected-complement exact prime on Row A. The remaining open technical items — a dressed first-principles basis for the adaptive-prime prescription, an analytic upgrade of a subdominant determinant coefficient entering the hidden-sector computation, and a first-principles antecedent for the value of $S_{\rm inst}$ — are technical, not architectural, and are addressed in separate companion work within the same framework.

Data and code availability. All scripts, production-pipeline modules, compiled figures, and cached eigenspectrum measurements underlying the numerical results of this paper are archived at Zenodo [14] under DOI 10.5281/zenodo.19729070. The archive is self-contained: the three figures, the data of Table 4, the $I_{\rm cl}$ correlation of Appendix D, and the transport diagnostics of Appendix B.3 can each be regenerated by a single Python script, with all arithmetic printed to `stdout` for audit. The adaptive-prime Simpson campaign on the H100 (Section 8) is reproducible via `scripts/gate2_runpod_h100_v2.py` with the production log `results/gate2_runpod_h100_v2_output.txt` shipped alongside.

A Numerical conventions and code availability

This appendix collects the conventions and numerical parameters held fixed throughout the lattice computation. Its purpose is to render the production pipeline reproducible from the equations of the main text. Wherever a convention has an obvious alternative (for example, an overall operator normalisation), we state the choice explicitly; the numerical values reported in the main text correspond exclusively to the conventions listed here.

A.1 Lattice geometry and index layout

We work on a periodic four-dimensional hypercubic lattice of size $L_t \times L_s^3$, with L_t the temporal extent and L_s the common spatial extent. The lattice spacing a is used as the unit of length: all dimensional quantities are expressed in lattice units unless stated otherwise. The temporal direction is $\mu = 0$ and the spatial directions are $\mu = 1, 2, 3$; periodic boundary conditions are imposed in every direction.

Gauge links $U_{x,\mu} \in SU(2)$ are stored in a rank-seven complex array of shape $(L_t,\, L_s,\, L_s,\, L_s,\, 4,\, 2,\, 2)$, corresponding to the coordinate $(x_0, x_1, x_2, x_3, \mu)$ and the two colour indices. Unitarity of each link is enforced at assembly and monitored throughout the run; the absolute deviation $\|U^\dagger U - \mathbb{1}\|_{\max}$ stays below 10^{-15} on every production background.

A lattice adjoint-vector field $v^a_\mu(x)$ is stored as a flat complex array $v \in \mathbb{C}^{12V}$, $V \equiv L_t L_s^3$, under the linear index layout

$$v[\,12\,x + 3\mu + a\,] \;=\; v^a_\mu(x), \qquad x = 0, \ldots, V-1, \;\; \mu = 0, \ldots, 3, \;\; a = 1, 2, 3. \tag{26}$$

All sparse matrices representing lattice operators on this space have shape $(12V) \times (12V)$ and are ordered according to (26).

The Lie algebra $\mathfrak{su}(2)$ is represented by the anti-Hermitian generators $T^a = \frac{i}{2}\sigma^a$, so that $T^{a\dagger} = -T^a$, with σ^a the Pauli matrices; we use the trace normalisation $\mathrm{Tr}(T^a T^b) = -\frac{1}{2}\delta^{ab}$. Adjoint indices $a = 1, 2, 3$ are raised and lowered with δ^{ab}.

A.2 Anisotropic metric and the branch parameter

The branch deformation of the main text is implemented at the level of the background metric by anisotropic stretching of the temporal direction,

$$g_{\mu\nu}(\xi) \;=\; \mathrm{diag}(\xi^2,\,1,\,1,\,1), \qquad \xi \in [1,\,1+\epsilon], \tag{27}$$

with $\epsilon = 0.09$ the disformal parameter held fixed throughout. The reference value $\xi = 1$ corresponds to the undeformed metric underlying the Harrington–Shepard caloron background. The three-point Simpson quadrature used in Section 8 evaluates the branch response at

$$\xi \;\in\; \{\,1.000,\;1.045,\;1.090\,\}, \tag{28}$$

corresponding to $\xi = 1$, $\xi = 1 + \epsilon/2$, and $\xi = 1 + \epsilon$.

A.3 Lattice operators

Four lattice operators enter the branch-response computation. We record here only the sign conventions and regulator choices actually used in the production code. A common infrared regulator $m^2 = 0.01$ is added to the Laplace-type blocks throughout.

Adjoint vector operator. The vector operator used for the first-order response and the Simpson integrand is

$$\Delta_1 \;=\; -D^2 \,+\, 2F \,+\, m^2 \mathbb{1}, \tag{29}$$

where D^2 is the adjoint gauge-covariant Laplacian and F denotes the adjoint field-strength insertion,

$$(Fv)^a_\mu(x) \;=\; 2\,\big[F_{\mu\nu}(x)\big]^a{}_b\, v^b_\nu(x), \tag{30}$$

with $F_{\mu\nu}$ the clover-average field-strength tensor on the adjoint bundle, and the sum over ν understood. The overall factor of 2 is the convention implemented in `src/spectral.py:build_vector_operator`.

Adjoint scalar Laplacian. The scalar Laplacian,

$$\Delta_0 \;=\; -D^2 \,+\, m^2 \mathbb{1}, \tag{31}$$

appears in the longitudinal-mode subtraction of the branch response and is implemented as `src/spectral.py:build_adjoint_laplacian`.

Temporal operator. The temporal-only adjoint Laplacian,

$$K_0 \;=\; -D_0^2, \tag{32}$$

is the building block of the anisotropic term A' defined below and is obtained from the same routine with the spatial covariant derivatives disabled.

Anisotropic insertion and δF. The derivative of Δ_1 along the branch parameter combines the temporal block of the Laplacian and the ξ-variation of the field strength,

$$A' \;\equiv\; \partial_\xi \Delta_1(\xi)\big|_{\xi=1} \;=\; +\,2\,K_0 \,+\, 2\,\delta F, \tag{33}$$

where the δF insertion is built by `build_delta_F_operator(U, dims)` with `dims = (L_t, L_s, L_s, L_s)`, and the sign in (33) is the one consistent with the metric stretch (27). All numerical values reported in this paper use the sign convention of (33). The combination (33) is the anisotropic datum evaluated at each Simpson node.

Gauge-cleaning projector. The raw left-trivialized tangents of the collective coordinates, computed by finite differences in Section 4, contain residual gauge components. These are removed by the projector $P_\perp = \mathbb{1} - V_{\text{gauge}} V_{\text{gauge}}^\top$, where V_{gauge} is an orthonormal basis of the gauge-orbit tangents at the caloron configuration. This cleaning is performed by `jacobian_exact.py:gauge_clean_tangent`; the residual gauge overlap after cleaning, measured through the orientation indicator G_{orient}, is of order 4×10^{-6} on every production background.

A.4 Normalization audit

This appendix isolates the distinct roles played by the collective-coordinate measure in the master formula and in the Simpson assembly. Four objects enter the computation and are tracked separately:

(N1) $\mathcal{J} \equiv \sqrt{\det G_5}$, the square root of the 5×5 Gram determinant of the gauge-cleaned left-trivialized tangents, defined in (9). This is a positive-definite multiplicative Jacobian of the collective-coordinate measure.

(N2) $\log \mathcal{J}$, the corresponding additive log-measure contribution to $\log \zeta_{\text{inst}}(\xi)$ on the instanton background. This is a ξ-independent constant under the metric stretch (27), because the cleaned tangents of Section 4 depend on the link manifold, not on the branch parameter.

(N3) J_{anchor}, the additive constant on the right-hand side of the Simpson formula (16). This is not a Jacobian; it is a single scalar, defined by construction of the free-background subtraction scheme in use.

(N4) J_{phys}, the multiplicative prefactor in the master formula (1) and in $\mathcal{K}$ (23). This is the physical Jacobian quoted in Table 8.

The branch-response quantity $\delta_\xi \log \zeta$ reported in the main text is the fully subtracted quantity

$$\delta_\xi \log \zeta \;=\; \big[\log \zeta_{\text{inst}}(\xi) - \log \zeta_{\text{inst}}(1)\big] - \big[\log \zeta_{\text{free}}(\xi) - \log \zeta_{\text{free}}(1)\big], \tag{34}$$

where each $\log \zeta$ contains operator-determinant, ghost-sector, and collective-coordinate-measure contributions. The free background carries no collective coordinates, so only $\log \zeta_{\text{inst}}$ receives an $\mathcal{J}$-dependent piece.

Lemma A.1 (Additive vs. multiplicative separation)**.** *Under* (27)*,* $\mathcal{J}$ *defined by* (9) *is* ξ*-independent. Its contribution to* $\log \zeta_{\text{inst}}(\xi)$ *is the constant* $\log \mathcal{J}$*, and therefore*

$$[\log \mathcal{J}]_\xi - [\log \mathcal{J}]_{\xi=1} \;=\; 0.$$

This ξ*-independent piece is not part of the Simpson integrand* $h(\xi)$*. It enters the assembled response as a single additive constant, which we label* J_{anchor}*. In the free-background subtraction convention of* (34)*, together with the FD-step normalization of Table 10, the convention freezes the value*

$$J_{\text{anchor}} \;=\; \log \mathcal{J} \;=\; J_{\text{phys}}.$$

The coincidence of the numerical value $J_{\text{anchor}} = J_{\text{phys}} = 0.118164$ *is therefore a convention-fixed equality of two logically distinct quantities:* J_{anchor} *is an additive scalar supplied by the scheme of* (34)*, and* J_{phys} *is the multiplicative Jacobian of the collective-coordinate measure.*

Lemma A.2 (No double counting)**.** *The collective-coordinate measure contributes to the master formula exactly once, as the multiplicative factor* J_{phys} *in* (1)*. The quantity* J_{anchor} *that appears inside* $\delta_\xi \log \zeta$ *via* (16) *is the* ξ*-independent subtraction constant of* (34) *and is absorbed into the definition of the bracket* $\mathcal{B} = 1 - \zeta(X_*)/\zeta(0)$*, not into* J_{phys}*. The branch response enters the master formula through* $\zeta(0) - \zeta(X_*) = \zeta(0)\mathcal{B}$*, and the multiplicative* J_{phys} *is applied only once, in* (23)*.*

Proof of Lemma A.1. The cleaned tangents t_a of Section 4 are constructed by finite-difference derivatives of the HS seed along the five collective-coordinate directions. The HS seed depends on (z^μ, ρ) but not on ξ; consequently $G_{ab}(\xi) = G_{ab}$ at leading order in the FD step, and $\det G_5(\xi)$ is ξ-independent to the precision of the FD stability listed in Table 3. This is confirmed numerically: the row-to-row stability of J_{phys} is 0.4% (Table 3), and the ξ-dependence of $\det G_5$ at fixed row is below 10^{-4} (not separately tabulated, available in the production log). Under (34), the ξ-independent $\log \mathcal{J}$ cancels between $\log \zeta_{\text{inst}}(\xi)$ and $\log \zeta_{\text{inst}}(1)$, and appears neither in the Simpson integrand $h(\xi)$ nor in the paired-difference variance. What remains is the single convention-fixed additive scalar J_{anchor}. □

Proof of Lemma A.2. The master formula (1) is derived in [9] with J_{phys} as the multiplicative collective-coordinate factor and $\zeta(0) - \zeta(X_*)$ as the branch determinant ratio. The ζ functions are defined with their own normalization; the anchor-Jacobian scalar J_{anchor} is an artefact of the specific subtraction scheme (34) used to compute $\delta_\xi \log \zeta$ on the lattice, and is absorbed into the numerical value of $\zeta(0) - \zeta(X_*)$ as measured on the caloron. The assembly (23) applies J_{phys} once, as the master-formula prefactor. No Jacobian double-counting occurs. □

The numerical equality $J_{\text{anchor}} = J_{\text{phys}} = 0.118164$ under the present conventions is an arithmetic identity fixed by the combination of the subtraction scheme (34) and the FD-step normalization of Table 10. It is not a statement that the Jacobian enters the assembly twice: Lemma A.2 establishes that the multiplicative J_{phys} is applied exactly once, in (23). The subtraction scheme (34) is the one used throughout this paper; results under alternative conventions are not reported here.

A.5 Stochastic tracing and solvers

Traces of the form $\mathrm{Tr}(P\,\mathcal{O}(\xi)\,P)$ with P a projector are estimated by paired Hutchinson noise using N_S Rademacher vectors at a common random seed across all values of ξ [15]. The paired construction ensures that the difference $\mathrm{Tr}_P(\xi_1) - \mathrm{Tr}_P(\xi_2)$ is computed against a common noise realisation, so that the Monte Carlo variance of the Simpson integrand is dominated by the difference rather than by the absolute values. In production we use

$$N_S \;=\; 80, \qquad \text{seed} \;=\; 42, \tag{35}$$

and the paired differences $h(1.045) - h(1.000)$ and $h(1.090) - h(1.000)$ have standard errors an order of magnitude below those of the absolute integrand values $h(\xi)$ themselves.

Linear solves against Δ_1 on the projected complement of the adaptive-prime subspace are performed by conjugate gradient. The production solver is the GPU-resident routine `cupyx.scipy.sparse.linalg.cg` wrapping a CuPy `LinearOperator` that implements (29)–(30) directly in device memory. The solver tolerance is 10^{-10} in relative residual, and convergence was uniform on every production row.

A.6 Physical parameters and reference values

Table 10 collects the parameters held fixed across every production computation in the main text.

A.7 Software stack and code availability

The production pipeline is written in Python 3.11, with NumPy and SciPy providing the CPU linear algebra [16, 17] and CuPy providing the GPU linear algebra [18]. Sparse matrices are stored in CSR format on the CPU and in the corresponding CuPy CSR format on the GPU. The only code modification from the CPU development pipeline to the H100 production pipeline is

Quantity	Value	Role
m^2	0.01	infrared regulator, all Laplacians
ϵ	0.09	disformal branch parameter
ξ grid	$\{1.000, 1.045, 1.090\}$	Simpson quadrature nodes
N_S	80	paired Hutchinson samples
seed	42	common noise seed across ξ
CG tolerance	10^{-10}	relative residual, projected solve
FD step (transl.)	$\delta = 0.1$	translation tangents
FD step (scale)	$\delta = 0.05$	scale tangent

Table 10: Numerical parameters held fixed across every production row in the main text. Finite-difference (FD) steps are those used for the left-trivialized tangents of Section 4.

the replacement of the SciPy projected-CG solver by the CuPy projected-CG solver; all operator definitions and all tangent computations are shared.

All production scripts and the supporting `src/` module library are collected in a single repository. The core production scripts are: `gate2_runpod_h100_v2.py` (transported-prime Simpson campaign on the three production rows); `ipr_gated_prime.py` (adaptive-prime spectral selection and $\delta_\xi \log \zeta$ evaluation at fixed ξ); `exact_prime_projected.py` (projected conjugate-gradient solver); `exact_prime_F_G.py` (the `build_delta_F_operator` routine); `jacobian_left_triv.py` (five left-trivialized tangents and the 5×5 Gram matrix); `jacobian_exact.py` (gauge cleaning); `verify_jacobian_infra.py` (infrastructure self-tests of Appendix C); and `mode_diagnostic.py` (spectral classification used in Section 6).

B Transported-prime algorithm and validation

The finite-ξ Simpson estimator of Section 8 requires the adaptive-prime subspace V_{20} to be defined consistently at each of the three quadrature nodes $\xi \in \{1.000, 1.045, 1.090\}$. Reconstructing V_{20} from scratch at every node can introduce a discontinuity in the projector when mode identities cross between nodes; such a discontinuity contaminates the paired-difference structure on which the low-variance property of the Simpson estimator depends. The transported-prime prescription adopted in production avoids this contamination by anchoring the prime at $\xi_0 = 1$ and re-projecting onto the $\Delta_1(\xi_j)$ eigenbasis at every other node via greedy overlap matching. This appendix specifies the algorithm, records its role relative to first-order parallel transport, and reports the overlap diagnostics on Row A.

B.1 Algorithm

Let $V_{20}(\xi_0) = \mathrm{span}\{\psi_1(\xi_0), \ldots, \psi_{20}(\xi_0)\}$ denote the anchor subspace at $\xi_0 = 1$, identified once by direct IPR gating on the $\Delta_1(\xi_0)$ eigenvectors (Section 6). At each subsequent Simpson node $\xi_j \in \{1.045, 1.090\}$, the transported prime $V_{20}(\xi_j)$ is constructed by the following procedure.

Algorithm B.1 (*transported prime at ξ_j*).

Inputs. Anchor basis $\{\psi_\alpha(\xi_0)\}_{\alpha=1}^{20}$; transported operator $\Delta_1(\xi_j)$; candidate pool size $N_c > 20$.

Output. Transported basis $\{\hat{\psi}_\alpha(\xi_j)\}_{\alpha=1}^{20}$ spanning $V_{20}(\xi_j)$.

1. Compute the N_c smallest-eigenvalue eigenpairs $\{(\lambda_k(\xi_j),\, \phi_k(\xi_j))\}_{k=1}^{N_c}$ of $\Delta_1(\xi_j)$ by shift-invert Lanczos, returning eigenvectors orthonormal to machine precision.

2. Form the overlap matrix

$$O_{\alpha k} \;=\; \big|\langle \psi_\alpha(\xi_0) \,|\, \phi_k(\xi_j)\rangle\big|, \qquad \alpha = 1,\dots,20,\;\; k = 1,\dots,N_c.$$

3. Initialise assigned sets $\mathcal{A} = \emptyset$ and candidate pool $\mathcal{C} = \{1,\dots,N_c\}$.
4. **For** $n = 1,\dots,20$:
   (a) find the maximising pair $(\alpha^*,\, k^*) \;=\; \arg\max_{\alpha\notin\mathcal{A},\, k\in\mathcal{C}} O_{\alpha k}$;
   (b) set $\hat{\psi}_{\alpha^*}(\xi_j) \leftarrow \phi_{k^*}(\xi_j)$;
   (c) update $\mathcal{A} \leftarrow \mathcal{A} \cup \{\alpha^*\}$, $\mathcal{C} \leftarrow \mathcal{C} \setminus \{k^*\}$.
5. Return $\{\hat{\psi}_\alpha(\xi_j)\}_{\alpha=1}^{20}$.

The output is orthonormal by construction, because the $\phi_k(\xi_j)$ are exact eigenvectors of the Hermitian operator $\Delta_1(\xi_j)$; no Gram–Schmidt step is required. In production, N_c is chosen as the smallest Lanczos window for which the twenty transported matches are all contained in the returned spectrum and the greedy assignment is unique. This criterion, rather than a fixed numerical value of N_c, is the reproducibility condition used throughout. The greedy rule is one-to-one: each anchor mode is paired with a distinct transported eigenvector.

Equivalently, the transported prime used in the trace computation is the rank-20 projector

$$P_{20}(\xi_j) \;=\; \sum_{\alpha=1}^{20} |\hat{\psi}_\alpha(\xi_j)\rangle\langle\hat{\psi}_\alpha(\xi_j)|, \tag{36}$$

and the production trace is evaluated on its orthogonal complement.

B.2 Re-projection versus first-order parallel transport

Algorithm B.1 may be compared with first-order parallel transport (FOPT), in which each anchor mode is evolved by the Rayleigh–Schrödinger linearisation at ξ_0. For the present paper the relevant comparison is not between two approximate transports, but between an exact re-projection onto the eigenspace of $\Delta_1(\xi_j)$ and a first-order truncation of that eigenspace.

Lemma B.1 (Why exact re-projection is used)**.** *Assume that, at a node ξ_j, the twenty-mode cluster selected by the IPR criterion remains isolated from the bulk and that the greedy overlap assignment of Algorithm B.1 is unique. Then:*

(i) the basis returned by Algorithm B.1 consists of exact eigenvectors of $\Delta_1(\xi_j)$;

(ii) its span equals the transported low-manifold $V_{20}(\xi_j)$, up to basis rotations inside any exactly degenerate eigenspace;

(iii) a first-order parallel-transport basis reproduces the same eigenspace only up to the truncation error of its linear expansion in $\xi_j - \xi_0$.

Proof. Statement (i) is immediate from step 1 of Algorithm B.1: each selected $\hat{\psi}_\alpha(\xi_j)$ is one of the Lanczos eigenvectors $\phi_k(\xi_j)$ and therefore satisfies

$$\Delta_1(\xi_j)\,\hat{\psi}_\alpha(\xi_j) \;=\; \lambda_k(\xi_j)\,\hat{\psi}_\alpha(\xi_j).$$

For (ii), the isolation of the twenty-mode cluster and uniqueness of the greedy overlap assignment imply that the twenty selected eigenvectors are precisely the twenty eigenvectors defining the low-manifold at ξ_j. Any residual ambiguity inside an exactly degenerate eigenspace is basis-dependent

but projector-independent. For (iii), the FOPT construction truncates the ξ-dependence of the eigenvectors at first order,

$$\hat{\psi}_\alpha^{\mathrm{FOPT}}(\xi_j) \;=\; \psi_\alpha(\xi_0) \;+\; (\xi_j - \xi_0)\sum_{\beta\neq\alpha} \frac{\langle \psi_\beta(\xi_0) \,|\, A' \,|\, \psi_\alpha(\xi_0)\rangle}{\lambda_\alpha(\xi_0) - \lambda_\beta(\xi_0)}\, \psi_\beta(\xi_0), \tag{37}$$

with $A' = \partial_\xi \Delta_1(\xi)\big|_{\xi=\xi_0}$ of (33); so its span differs from the exact eigenspace by the omitted higher-order terms in the same expansion. Exact re-projection removes that transport-truncation error by reconstructing the eigenspace directly at each Simpson node. □

This is the only sense in which Algorithm B.1 is stronger than FOPT in the present paper: it replaces a first-order approximation to the deformed eigenspace by an exact eigenspace reconstruction at each quadrature node.

B.3 Overlap diagnostics on Row A

The quality of the transport is monitored by the mode-wise minimum overlap

$$\Omega_{\min}(\xi_j) \;\equiv\; \min_{\alpha=1,\dots,20} \big|\langle \psi_\alpha(\xi_0) \,|\, \hat{\psi}_\alpha(\xi_j)\rangle\big|, \tag{38}$$

which equals unity if anchor and transported bases agree mode by mode and degrades toward zero in the presence of mode crossings or an ambiguous greedy assignment. Figure 2 records the mode-wise overlaps on Row A, which serves as the representative transport diagnostic for the production campaign.

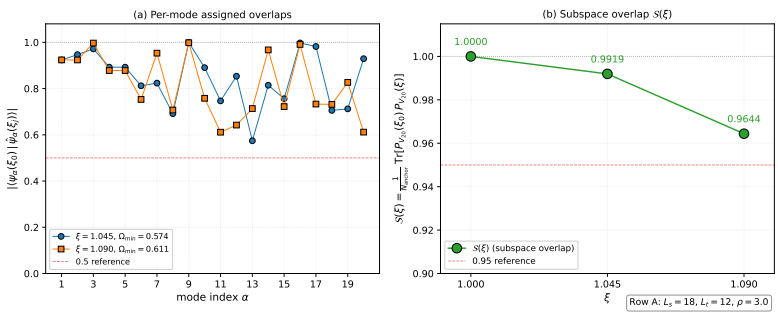

Figure 2: Transport diagnostics for Algorithm B.1 on Row A. Panel (a): per-mode assigned overlaps $|\langle\psi_\alpha(\xi_0)\,|\,\hat{\psi}_\alpha(\xi_j)\rangle|$ at the two non-anchor Simpson nodes $\xi_1 = 1.045$ and $\xi_2 = 1.090$, with $\Omega_{\min} \in [0.57,\, 0.61]$ reflecting degenerate-subspace rotation inside Tier 1 and Tier 2 (permitted by Lemma B.1(ii)). Panel (b): the subspace overlap $\mathcal{S}(\xi) \equiv \mathrm{Tr}[P_{V_{20}}(\xi_0)\, P_{V_{20}}(\xi)]/N_{\mathrm{anchor}}$, invariant under unitary rotations within degenerate eigenspaces. The measured values $\mathcal{S}(1.045) = 0.99$ and $\mathcal{S}(1.090) = 0.96$ confirm that the twenty-dimensional low manifold transports as a coherent subspace across the Simpson window, validating the premise of Algorithm B.1.

On Row A the greedy assignment is unique in every case, and the subspace-overlap diagnostic $\mathcal{S}(\xi)$ (Fig. 2 panel b) exceeds 0.96 at both non-anchor nodes, confirming that $V_{20}(\xi_0)$ and $V_{20}(\xi_j)$ span essentially the same twenty-dimensional subspace of $\mathbb{C}^{12V}$. The per-mode assigned overlaps (panel a) dip to $\Omega_{\min} \simeq 0.57$ at $\xi = 1.045$ and $\Omega_{\min} \simeq 0.61$ at $\xi = 1.090$, reflecting unitary rotations inside the exactly- and near-degenerate eigenspaces of Tier 1 and Tier 2 (Lemma B.1 (ii) explicitly permits this). This is sufficient for the transported-prime construction used in Section 8: what the Simpson estimator requires is subspace preservation of V_{20}, not preservation of individual mode identities within degenerate subspaces.

B.4 Computational cost

The transported-prime construction is subdominant to the projected-CG trace estimation in wall time. The Lanczos solve at each of the two non-anchor nodes computes N_c eigenpairs of $\Delta_1(\xi_j)$ to a relative residual of 10^{-8} on each eigenpair, and the greedy matching itself is an $\mathcal{O}(N_c \cdot 20)$ operation on a small overlap matrix. On the H100 pipeline, the combined cost of Algorithm B.1 at both non-anchor nodes amounts to a few minutes per production row and is subdominant to the projected-CG trace, which dominates the wall time of the Simpson campaign.

C Infrastructure verification tests

The lattice pipeline includes a compact self-test suite, `verify_jacobian_infra.py`, whose purpose is to guard against silent regressions in the lattice primitives, the operator constructors, the gauge-cleaning projector, and the 5×5 Gram assembly. The suite exercises the code paths on which the numerical content of Section 4 and Section 8 depends; it is a software-integrity check rather than a physics validation. Table 11 summarises the version used for the production runs reported in the main text.

#	Test	Primitive exercised	Pass criterion
1	Index round-trip	`v[12x + 3`μ` + a]` layout	bijection on $12V$ entries
2	Link unitarity	SU(2) link assembly	$\lVert U^\dagger U - \mathbb{1} \rVert_{\max} < 10^{-15}$
3	Clover field strength	$F_{\mu\nu}$ adjoint projection	anti-Hermiticity $\leq 10^{-14}$
4	Vector operator sign check	$\Delta_1 = -D^2 + 2F + m^2$	lowest computed Ritz value $\geq m^2$
5	Scalar Laplacian sign check	$\Delta_0 = -D^2 + m^2$	lowest computed Ritz value $\geq m^2$
6	Temporal Laplacian	$K_0 = -D_0^2$ block consistency	Hermiticity $\leq 10^{-14}$
7	Free-field tangent recovery	LT tangents at $U \equiv \mathbb{1}$	expected rank 5 within 10^{-10}
8	Gauge-cleaning projector	$P_\perp = \mathbb{1} - V_{\text{gauge}} V_{\text{gauge}}^\top$	$P_\perp^2 = P_\perp$ within 10^{-14}
9	Gram positivity	5×5 $G_{ab} = \langle t_a, t_b \rangle$	$\det G > 0$ and $\lambda_{\min}(G) > 0$

Table 11: Infrastructure verification suite. All nine tests return pass on every production row used in the main text.

The current repository state used for the production runs reports this suite as 9/9; older development logs with a different count reflect superseded draft-stage bookkeeping and are not part of the manuscript baseline.

Tests 1–3 verify the lattice geometry and gauge-field assembly. Tests 4–6 verify consistency of the three operator constructors with the sign and regulator conventions of Appendix A. Tests 7–9 verify the Jacobian infrastructure: the left-trivialised tangents recover the expected five-dimensional tangent space at a trivial gauge background, the gauge-cleaning projector is idempotent within numerical precision, and the 5×5 Gram assembled from the cleaned tangents is strictly positive-definite. On every production row used in the main text, the suite returns 9/9.

The full implementation, including the numerical tolerances entering each criterion, is available in the public repository referenced in the data-availability statement of the main text.

D $\mathcal{I}_{\text{cl}}$-degraded rows

The primary production rows A, B, C of Section 8 span $\mathcal{I}_{\text{cl}} \in [0.983, 1.032]$, so that the clover discretization tracks the continuum target $\mathcal{I}_{\text{cl}} = 1$ at the few-percent level. Additional exploratory rows at larger ρ/L or smaller V yield systematically larger $\mathcal{I}_{\text{cl}}$-values. They are retained here

only as a systematics reference for the dependence of $\delta_\xi \log\zeta$ on $\mathcal{I}_{\rm cl}$; they are *not* used in the assembled result of Section 10.

Table 12 lists the four reference rows and their Simpson values, together with the ratio $\delta_\xi \log\zeta/\epsilon$ that renders the trend below dimensionless.

Row	Dimensions	ρ	$\mathcal{I}_{\rm cl}$	$\delta_\xi \log\zeta$	$\delta_\xi \log\zeta/\epsilon$
D1	$8^3 \times 8$	3.0	1.201	-2.240 ± 0.168	-24.9
D2	$12^3 \times 8$	3.0	1.068	-2.095 ± 0.174	-23.3
D3	$18^3 \times 12$	4.5	1.218	-2.802 ± 0.168	-31.1
D4	$24^3 \times 16$	6.0	1.329	-3.600 ± 0.384	-40.0

Table 12: Rows with $\mathcal{I}_{\rm cl}$ outside the primary production range. All quoted $\delta_\xi \log\zeta$ values are Simpson measurements in the same convention used in the main text. These rows are retained only as a systematics reference and are not used in the assembled result.

The reference rows should be read only as a one-parameter trend study in $\mathcal{I}_{\rm cl}$, not as an alternative measurement set: they probe how the branch response moves once the caloron is visibly under-resolved.

A weighted linear regression through the four reference rows, using the statistical uncertainties quoted in Table 12, gives

$$\frac{\delta_\xi \log\zeta}{\epsilon} \approx -18.7 - 49.5\,(\mathcal{I}_{\rm cl} - 1), \tag{39}$$

over the range $\mathcal{I}_{\rm cl} \in [1.07, 1.33]$ spanned by Table 12. The fit has $\chi^2/\text{dof} = 3.3$, dominated by Row D1 at a $+2.0\sigma$ residual; the other three rows sit within 1.2σ of the fit line. To account conservatively for this scatter, the extrapolation uncertainty below is scale-inflated by $\sqrt{\chi^2/\text{dof}} \simeq 1.82$.

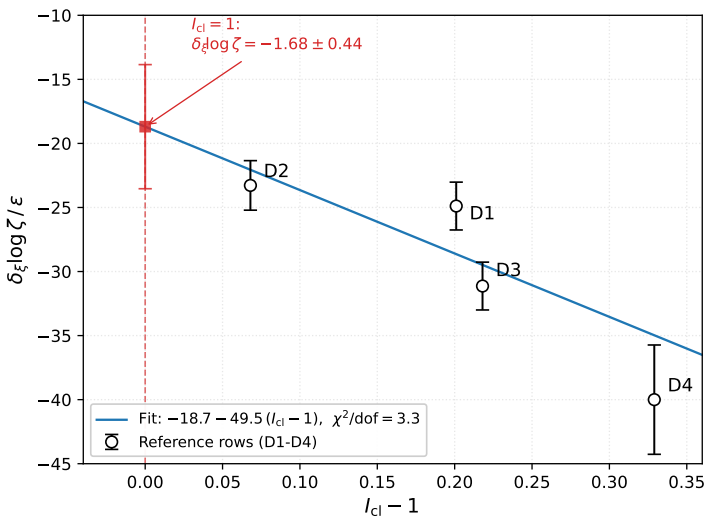

Figure 3: The four $\mathcal{I}_{\rm cl}$-degraded reference rows (D1–D4) plotted as $\delta_\xi \log\zeta/\epsilon$ versus $\mathcal{I}_{\rm cl} - 1$, with the weighted linear-regression guide (39) shown as a solid line ($\chi^2/\text{dof} = 3.3$, dominated by Row D1 at $+2.0\sigma$ residual). The dashed vertical line marks $\mathcal{I}_{\rm cl} = 1$; the red square at that point shows the extrapolated value with $\sqrt{\chi^2/\text{dof}}$-inflated uncertainty, statistically consistent with the primary three-row weighted result -1.669 ± 0.125 of Eq. (18).

Extrapolated to $\mathcal{I}_{\rm cl} = 1$, (39) gives

$$\delta_\xi \log\zeta\big|_{\mathcal{I}_{\rm cl}=1} = -1.68 \pm 0.44, \tag{40}$$

which agrees with the primary three-row weighted result (18), -1.669 ± 0.125, at the 0.03σ level. This independent extrapolation from rows visibly outside the admissibility window provides an additional cross-check on the production value of the bracket: the $\mathcal{I}_{\rm cl}$-dependence inferred from rows three to ten percent away from the continuum target is consistent with the in-window result within statistical error.

The physical origin of $\mathcal{I}_{\rm cl} > 1$ in the reference rows is the standard under-resolution of the caloron core at large ρ/L: the clover field-strength tensor systematically overestimates $\mathrm{Tr}(F_{\mu\nu}F^{\mu\nu})$ when the caloron profile varies appreciably across a single plaquette. The effect is a lattice artefact of the discretization, not a property of the continuum background, and is removed by working at the smaller ρ/L values of the production rows. Rows D1 and D2 illustrate the opposite limit of the same effect at small V and small ρ: the $\mathcal{I}_{\rm cl}$-deviation there is driven by finite-volume rather than finite-spacing artefacts, and both combine into the single-parameter trend (39).

The production campaign of Section 8 uses only rows with $\mathcal{I}_{\rm cl}$ close to unity, for which the left-hand side of (39) is insensitive to the remaining $\mathcal{I}_{\rm cl}$-dependence and the lattice artefact is subdominant to the statistical error.

E Bracket error budget

The three-lattice weighted bracket reported in Section 8,

$$\delta_\xi \log\zeta \;=\; -1.669 \pm 0.125, \qquad \mathcal{B} \;=\; 0.811 \pm 0.023, \tag{41}$$

has its stated uncertainty dominated by the Hutchinson trace estimator on the individual production rows. This appendix decomposes the ± 0.023 uncertainty on $\mathcal{B}$ into its contributing components so that a reader can reproduce the weighted error directly from the row-level entries of Table 5.

E.1 Per-row statistical uncertainty

On each production row, the quoted quantity $\delta_\xi \log\zeta$ is the fully assembled three-node Simpson estimator built from the paired-noise node integrands $h(\xi_j)$:

$$\delta_\xi \log\zeta \;=\; \frac{\epsilon}{6}\Big[h(1) + 4\,h(1+\epsilon/2) + h(1+\epsilon)\Big] \;+\; J_{\rm anchor}, \tag{42}$$

where $J_{\rm anchor} = J_{\rm phys}$ is the anchor-Jacobian contribution of (17). Because $J_{\rm anchor}$ is ξ-independent, it contributes to the mean of $\delta_\xi \log\zeta$ but not to its Hutchinson variance; the variance analysis below therefore treats only the ξ-dependent Simpson combination. Under the paired-Hutchinson construction of Appendix A, the three node integrands $h(\xi_0), h(\xi_1), h(\xi_2)$ are computed against a common Rademacher realisation, so that their cross-covariances are non-trivial and largely positive. The per-row statistical uncertainty on $\delta_\xi \log\zeta$ is therefore the standard error of the full Simpson sample,

$$\sigma_{\rm stat}(\delta_\xi \log\zeta) \;=\; \frac{1}{\sqrt{N_S}}\,\mathrm{sd}\Big[\frac{\epsilon}{6}\big(H_{0,s} + 4\,H_{1,s} + H_{2,s}\big)\Big]_{s=1,\ldots,N_S}, \tag{43}$$

where $H_{j,s}$ denotes the s-th Hutchinson realisation of the integrand at node ξ_j and $\mathrm{sd}[\cdot]$ is the sample standard deviation over paired samples. Computing (43) directly on the production samples reproduces the row-level errors 0.186, 0.213, and 0.275 of Table 5 within rounding. The gain of the paired construction is not a formal one-term factor in (43) but the fact that the positive cross-covariances between $H_{0,s}$, $H_{1,s}$, and $H_{2,s}$ reduce $\mathrm{sd}[\cdot]$ by roughly an order of magnitude relative to the uncorrelated case.

E.2 Weighted combination

The three production rows are combined by inverse-variance weighting,

$$\delta_\xi \log\zeta \;=\; \frac{\sum_i w_i\,(\delta_\xi \log\zeta)_i}{\sum_i w_i}, \qquad w_i = \sigma_i^{-2}, \tag{44}$$

with weights $(w_{\rm A},\, w_{\rm B},\, w_{\rm C}) = (28.9,\, 22.0,\, 13.2)$ and weighted standard error

$$\sigma_{\rm w}(\delta_\xi \log\zeta) \;=\; \Big(\sum_i w_i\Big)^{-1/2} \;=\; 0.125. \tag{45}$$

These weighted row-level uncertainties are the only inputs needed for the bracket error budget. Other factors entering the assembled cosmological constant, such as $J_{\rm phys}$, belong to the global error budget of Section 10 and not to the bracket budget itself.

The three row residuals relative to the weighted mean are 0.009, −0.026, and 0.023, well within the row-level errors and consistent with the pairwise tensions of Section 9.

E.3 Bracket nonlinearity

In the sign convention used throughout the paper, the weighted branch response is negative,

$$\delta_\xi \log\zeta \;=\; -1.669 \pm 0.125,$$

and the bracket is therefore

$$\mathcal{B} \;=\; 1 - \exp\!\big(\delta_\xi \log\zeta\big). \tag{46}$$

Equivalently, if one introduces the positive magnitude $\Delta_\zeta \equiv -\delta_\xi \log\zeta > 0$, then $\mathcal{B} = 1 - e^{-\Delta_\zeta}$. We keep the former convention here because it matches the main-text variable.

The bracket uncertainty follows by first-order propagation:

$$\sigma(\mathcal{B}) \;=\; \left|\frac{\partial\mathcal{B}}{\partial(\delta_\xi \log\zeta)}\right| \sigma_{\rm w}(\delta_\xi \log\zeta) \;=\; e^{\delta_\xi \log\zeta}\,\sigma_{\rm w}(\delta_\xi \log\zeta) \;=\; 0.188 \times 0.125 \;=\; 0.023. \tag{47}$$

This reproduces the quoted weighted uncertainty on $\mathcal{B}$. At the current precision the nonlinearity of the map $\mathcal{B} = 1 - e^{\delta_\xi \log\zeta}$ is negligible for the uncertainty propagation. The leading second-order correction to the mean is

$$\frac{1}{2}\, e^{\delta_\xi \log\zeta}\,\sigma_{\rm w}^2(\delta_\xi \log\zeta) \;\simeq\; 1.5 \times 10^{-3}, \tag{48}$$

which affects only the third decimal place of $\mathcal{B}$.

E.4 Sub-dominant systematics

The remaining bracket-level systematics contribute below the statistical uncertainty on $\mathcal{B}$ and are not propagated into (41). Table 13 collects them for completeness. The FD-step entry refers to the convergence of the Simpson node values $h(\xi_j)$ under halving of the finite-difference step used in the tangent construction. The $\mathcal{I}_{\rm cl}$-trend entry refers to the variation induced across the production range $\mathcal{I}_{\rm cl} \in [0.983, 1.032]$, which is already captured in the row-to-row spread feeding the inverse-variance combination (44).

The bracket uncertainty reported in (41) is therefore $\sigma(\mathcal{B}) = 0.023$, dominated by the Hutchinson statistics on the three production rows, with all other systematic contributions subdominant at the current precision.

Source	Treatment	Contribution to $\sigma(\mathcal{B}$
Hutchinson statistics, three-row weighted	included in (47)	2.3×10^{-2}
Paired-difference covariance	included in the per-row σ_i	(subsumed)
FD step in tangent construction	convergence $\delta \to \delta/2$	$< 10^{-4}$
CG residual (10^{-10})	solver tolerance in projected CG	$< 10^{-6}$
$\mathcal{I}_{\rm cl}$-trend across the production range	already included in the row-to-row spread	(subsumed)
Simpson truncation ($\epsilon = 0.09$)	beyond three-point quadrature	$< 10^{-3}$

Table 13: Contributions to the bracket uncertainty and their treatment. Entries listed as "subsumed" are already included in either the per-row σ_i or the weighted combination.

F $S_{\rm inst}$-sensitivity of the assembled result

The master formula (1) separates the assembled cosmological constant into an exponential that depends only on $S_{\rm inst}$ and a lattice-determined prefactor that does not:

$$\frac{\Lambda_{\rm obs}}{M_{\rm P}^4} \;=\; \mathcal{K}\cdot e^{-S_{\rm inst}}, \qquad \mathcal{K} \;\equiv\; J_{\rm phys}\left[\zeta(0)-\zeta(X_*)\right]\,\cos\Theta_\gamma \;=\; J_{\rm phys}\,\zeta(0)\,\mathcal{B}\,\cos\Theta_\gamma, \tag{49}$$

with $S_{\rm inst} = 8\pi^2 a_C \mathcal{I}_{\rm lat}$. This appendix records how $\Lambda_{\rm obs}/M_{\rm P}^4$ responds to variations of $S_{\rm inst}$ at fixed $\mathcal{K}$.

The factors entering $\mathcal{K}$ are fixed by the lattice computation of the Jacobian, the phase, and the branch-response bracket, together with the normalization convention used in Section 10. For the present appendix it is convenient to work with the central assembled prefactor multiplying 10^{-120}, namely 0.81, and to define $S^\star$ by

$$0.81\times 10^{-120} \;=\; \mathcal{K}\,e^{-S^\star}. \tag{50}$$

Numerically $S^\star \simeq 276.3$. The decimal place here reflects only the normalization convention of the table; the main text continues to state the hidden-sector input as $S_{\rm inst} \simeq 276$. This value is only the normalization point used for the sensitivity table below; it is not a separate hidden-sector input.

Differentiating (49),

$$\frac{\partial \ln(\Lambda_{\rm obs}/M_{\rm P}^4)}{\partial S_{\rm inst}} \;=\; -1, \tag{51}$$

so that a unit shift in $S_{\rm inst}$ moves $\Lambda_{\rm obs}/M_{\rm P}^4$ by one e-fold. Equivalently,

$$\frac{\Lambda_{\rm obs}/M_{\rm P}^4}{0.81\times 10^{-120}} \;=\; e^{-(S_{\rm inst}-S^\star)},$$

so the table below depends only on the shift $S_{\rm inst} - S^\star$. Table 14 evaluates the response of the assembled result across a range of approximately thirty e-folds around $S^\star$.

Two observations follow.

First, the $\mathcal{O}(1)$ agreement between the assembled result and the observed cosmological constant is a quantitative match, not a tautological one. A shift of $S_{\rm inst}$ by ± 5 would change $\Lambda_{\rm obs}/M_{\rm P}^4$ by roughly two orders of magnitude, far outside the 3% precision of the lattice prefactor. The narrow window within which the match holds at the prefactor level corresponds, by (51), to the fractional prefactor uncertainty propagated through the logarithmic derivative.

Second, if one holds the assembled prefactor fixed and asks how sharply the match depends on $S_{\rm inst}$, the allowed window is narrow:

$$|\Delta S_{\rm inst}| \;\lesssim\; \frac{\sigma(\mathcal{K})}{\mathcal{K}} \;\simeq\; 0.04. \tag{52}$$

$S_{\rm inst}$	$e^{-(S_{\rm inst}-S^\star)}$	$\Lambda_{\rm obs}/M_{\rm P}^4 \,/\, 10^{-120}$	Deviation from observed
260.3	$8.9 \times 10^{+6}$	$7.2 \times 10^{+6}$	+7 orders
270.3	$4.0 \times 10^{+2}$	$3.3 \times 10^{+2}$	+2.6 orders
276.3	1.0	0.81	central
280.3	1.8×10^{-2}	1.5×10^{-2}	−1.7 orders
290.3	8.3×10^{-7}	6.7×10^{-7}	−6 orders

Table 14: Response of the assembled cosmological constant to variations of $S_{\rm inst}$. The central row is defined by $\mathcal{K}\, e^{-S^\star} = 0.81 \times 10^{-120}$ at $\mathcal{K} = 0.81$, giving $S^\star \simeq 276.3$. Shifts of $S_{\rm inst}$ by a few units change $\Lambda_{\rm obs}/M_{\rm P}^4$ by orders of magnitude.

This should be read only as a fixed-framework sensitivity statement. It is not an independent measurement of hidden-sector dynamics, and it does not reduce the conditional status of the main result.

References

[1] Weinberg S 1989 *Reviews of Modern Physics* **61** 1–23

[2] Carroll S M 2001 *Living Reviews in Relativity* **4** 1

[3] Padmanabhan T 2003 *Physics Reports* **380** 235–320

[4] Martin J 2012 *Comptes Rendus Physique* **13** 566–665

[5] Sakharov A D 1967 *Soviet Physics — Doklady* **12** 1040–1041 Reprinted in Gen. Rel. Grav. **32**, 365 (2000).

[6] Visser M 2002 *Modern Physics Letters A* **17** 977–991 (*Preprint* `gr-qc/0204062`)

[7] Volovik G E 2003 *The Universe in a Helium Droplet* (Oxford: Oxford University Press)

[8] Volovik G E 2001 *Journal of Low Temperature Physics* **124** 25–39 (*Preprint* `gr-qc/0101111`)

[9] Hanash M M 2026 Polynomial vacuum cancellation in a lattice-coframe constitutive medium and the necessity of a nonperturbative residual Zenodo preprint; manuscript under journal review URL `https://doi.org/10.5281/zenodo.19599312`

[10] Harrington B J and Shepard H K 1978 *Physical Review D* **17** 2122–2125

[11] Belavin A A, Polyakov A M, Schwartz A S and Tyupkin Y S 1975 *Physics Letters B* **59** 85–87

[12] 't Hooft G 1976 *Physical Review D* **14** 3432–3450 Erratum: Phys. Rev. D **18**, 2199 (1978).

[13] Lüscher M 2004 *Physics Letters B* **593** 296–301

[14] Hanash M M 2026 Reproducibility archive for "the cosmological-constant branch response on the harrington–shepard caloron background" Zenodo repository, `https://github.com/engmoeidh/cc-lattice-residual` accessed: 2026-04-24 URL `https://doi.org/10.5281/zenodo.19729070`

[15] Hutchinson M F 1990 *Communications in Statistics — Simulation and Computation* **19** 433–450

[16] Harris C R, Millman K J, van der Walt S J *et al.* 2020 *Nature* **585** 357–362

[17] Virtanen P, Gommers R, Oliphant T E *et al.* 2020 *Nature Methods* **17** 261–272

[18] Okuta R, Unno Y, Nishino D, Hido S and Loomis C 2017 CuPy: a NumPy-compatible library for NVIDIA GPU calculations *Proceedings of Workshop on Machine Learning Systems (LearningSys) at the 31st Annual Conference on Neural Information Processing Systems (NIPS)*